Problem B

Probabilistic Prediction of Goal-Plane Crossings in Penalty Kicks from Two Early Ball Positions

Abstract

We use **two early ball positions**, measured 50 and 100 ms after release, to predict where a penalty crosses the goal plane. We weight aerodynamic trajectories by their agreement with both readings and follow each to the goal plane or ground. For the prescribed kick, centimetre rounding, exact frame times and steady wind up to 3 m/s rule out drag alone as the cause of the turn. Conditional on direct entry, the **most probable cell** is centred at $(2.56, 2.07)$ m. Six cells spanning 0.357 m^2 contain 90.5% of the conditional probability. Median arrival is 0.399 s after release.

At the common 100 ms cutoff, we scored prespecified calibration targets on a fixed rounded synthetic test and then matched observed hit counts. At 70/71 hits, mean region areas were 0.585 m^2 for two frames, 3.482 m^2 for the late frame and 1.222 m^2 for gravity. With tracking noise, gravity needed less area at 67/71 hits. An independent noisy test gave 343/368 hits overall and 90/110 for **stable-spin entries**. Across eight calibration batches, aerodynamics covered more kicks than gravity twice and fewer six times, with corresponding differences in area. Three match records lacked verified release times, leaving field accuracy unquantified.

Key innovations. A wind-adjusted test identifies bending that drag alone cannot explain. Two flights with identical rounded readings cross 0.922 m apart. Under the stated assumptions, any single-point forecast must miss at least one of those crossings by about **0.461 m** or more.

Contents

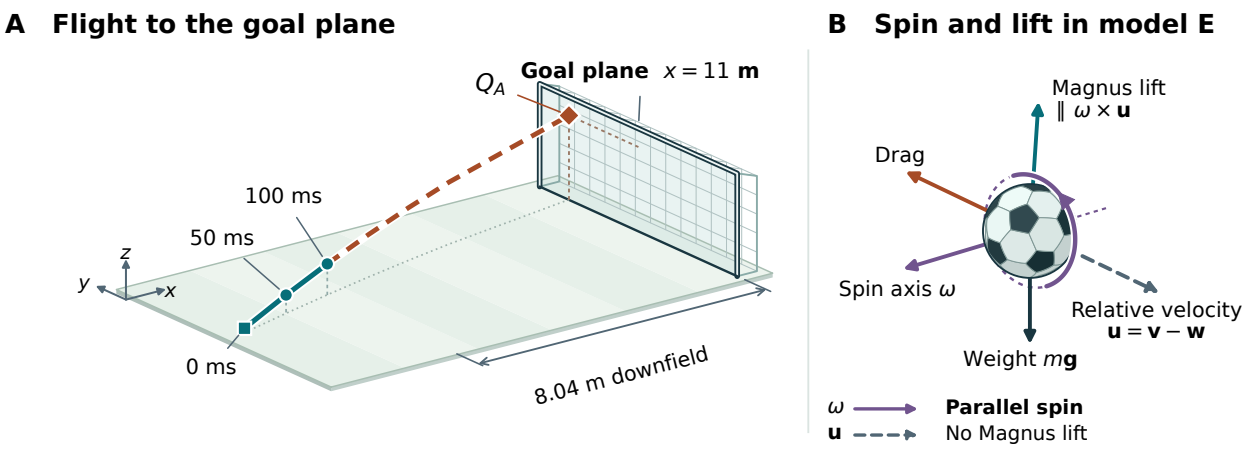
1 Introduction **3**
1.1 Problem Statement and Research Objectives . . . 3
1.2 Coordinates, Observations, and Outcomes . . . 3
1.3 Wind Threshold for Drag-Only Flight . . . 4
2 Flight Models and Prediction Method **5**
2.1 Aerodynamic Forces and Model Parameters . . . 5
2.2 Force and Time Scales . . . 5
2.3 Trajectory Weighting and Goal-Plane Mapping . . . 6
3 Results for the Prescribed Kick **7**
3.1 Most Probable Cell and Joint Prediction Region . . . 7
3.2 Constraints on Physical Parameters . . . 7
3.3 Sensitivity of the Most Probable Cell . . . 8
4 Limits of Two-Frame Prediction **8**
4.1 Ambiguity at Exact Coordinates . . . 8
4.2 Ambiguity under Centimetre Rounding . . . 9
5 Measurement Uncertainty **9**
5.1 Position, Timing, and Rounding Errors . . . 9
5.2 Sensitivity to Measurement Assumptions . . . 10
5.3 Combined Timing and Tracking Errors . . . 11
6 Forecast Validation at 100 ms **12**
6.1 Paired Forecasts at a Common Deadline . . . 12
6.2 Calibration of Prediction Regions . . . 12
6.3 Coverage under Centimetre Rounding . . . 13
6.4 Coverage under Tracking Noise . . . 13
6.5 Forecast Availability and Runtime . . . 14
6.6 Independent Stable-Spin Test . . . 15
6.7 Point and Event Forecast Scores . . . 16
6.8 Lift Priors and Calibration Variability . . . 17
7 Discussion and Limitations **18**
7.1 Field Evidence and Model Limitations . . . 18
7.2 Goalkeeper Decision Time . . . 18
8 Conclusion **19**
A Error Propagation at the Goal Plane **20**
B Synthetic Experiments and Forecast Robustness **22**
C Forecast Algorithm and Physical Checks **27**
D Timing of Additional Observations **29**
E Match Tracking and Requirements for Field Validation **31**
F Computational Reproducibility **32**
References **33**
Report on the Use of AI **33**

1 Introduction

1.1 Problem Statement and Research Objectives

A goalkeeper has two positions of the ball centre, measured 50 and 100 ms after a penalty leaves the mark. We seek the most likely point of direct entry through the goal plane and the uncertainty in that point, using only information available at the second reading [1]. We treat contact with the ground before the plane as a different outcome. For the prescribed kick, our highest-probability cell is centred at $(Y, Z) = (2.56, 2.07)$ m, about a metre inside the positive-y post.

By the second reading the ball has travelled 2.96 m toward the goal and has 8.04 m still to fly (Fig. 1). Its horizontal displacements show that the heading is turning toward smaller y relative to the first 50 ms chord. We first measure that turn, then ask which force histories could produce it and where their paths cross the goal plane. The prescribed kick has no observed endpoint, so we test the method on separate kicks (Table 1).



Release	50 ms	100 ms	Goal opening
(0.00, 0.00, 0.11) m	(1.50, 0.45, 0.44) m	(2.96, 0.86, 0.74) m	7.32 m wide × 2.44 m high

Figure 1. Two early readings and the role of spin.
A shows release (square), two ball centres (circles) and the model-S flight to Q_A (Sec. 4). Dashed segments join its stored positions, and the dotted line is its ground projection.
B uses a separate model-E release with prescribed spin and wind (Eq. (3)). The curved arrow gives the sense of rotation. Only the spin component perpendicular to the air-relative velocity contributes to Magnus lift. Arrow lengths are schematic.

1.2 Coordinates, Observations, and Outcomes

Let x point from the penalty spot toward the goal, y run across its face and z measure height above the ground. The ball centre starts at $\boldsymbol{r}_0 = (0, 0, 0.11)$ m. Its two prescribed positions, in metres, are

$$\boldsymbol{r}(0.050\,\mathrm{s}) = (1.50, 0.45, 0.44), \qquad \boldsymbol{r}(0.100\,\mathrm{s}) = (2.96, 0.86, 0.74). \tag{1}$$

For the reference forecast we take the release position and observation times as exact and interpret the displayed coordinates as rounded to the nearest centimetre. Each later coordinate q then represents $[q - \epsilon, q + \epsilon)$, with $\epsilon = 0.005$ m. The problem supplies no tracking accuracy, ball type, wind or spin. Section 5 examines how different observation assumptions affect this kick.

Table 1. Answers and evidence for the prescribed kick and the 100 ms prediction task.

Task	Main result	Evidence
Predict the crossing	Weight flights against the readings and propagate to the first event.	Sec. 2
Apply to the given kick	Modal cell (2.56, 2.07) m. Median arrival 0.399 s after release with exact timing.	Fig. 3
Quantify uncertainty	0.357 m^2 at 90.5% conditional model mass. For the same inputs, the worst-case point-error lower bound is about 0.461 m.	Figs. 3, 4
Test the method	At fixed 100 ms, region size and coverage depend on observation error and calibration.	Sec. 6

We follow the ball to its first forward crossing of $x = 11$ m or its first descent to one ball radius above ground. A direct entry G occurs when the centre crosses the 7.32 m wide opening with $-3.66 < Y < 3.66$ m and $0 < Z < 2.44$ m. The other outcomes are a crossing outside the opening, earlier ground contact, or neither event by 1.25 s. We estimate all four probabilities and the distribution of $Q = (Y, Z)$ conditional on G. This ball-centre convention omits frame collisions, rebounds and goalkeeper interventions.

1.3 Wind Threshold for Drag-Only Flight

The mean horizontal velocities over the first and second 50 ms are $(30.0, 9.0)$ and $(29.2, 8.2)$ m/s. Their directions differ. Under drag alone in a constant horizontal wind $\boldsymbol{w}$, the horizontal velocity keeps its direction relative to the air. The horizontal displacements $\boldsymbol{p}_i$ from release must therefore satisfy

$$\det(\boldsymbol{p}_1 - t_1\boldsymbol{w},\, \boldsymbol{p}_2 - t_2\boldsymbol{w}) = 0, \qquad \|\boldsymbol{w}\| \geq \frac{|\det(\boldsymbol{p}_1, \boldsymbol{p}_2)|}{\|t_2\boldsymbol{p}_1 - t_1\boldsymbol{p}_2\|}. \tag{2}$$

At the displayed coordinates, Eq. (2) requires at least 14.85 m/s. To allow for rounding, fix the wind first. The determinant is affine in each coordinate separately, so its value at any point of the four-dimensional box is a weighted average of its values at the 16 corners. All corners have the same sign at zero wind. The first possible zero as the wind disk expands is therefore a corner zero. Evaluating Eq. (2) at the corners gives an infimum of 4.353 m/s for the rounding intervals, above the assumed cap of 3 m/s. Collinearity is a necessary condition for a drag-only flight.

How large is the turn? With $h = 0.050$ s, the horizontal second difference $(\boldsymbol{p}_2 - 2\boldsymbol{p}_1)/h^2 = (-16, -16)$ m/s^2 is the net acceleration averaged over the first $2h$ with a triangular time weight. Projecting onto the left normal of the full air-relative chord gives -10.9 m/s^2 in still air at the displayed coordinates. The smallest magnitude over the allowed winds is about 8.92 m/s^2 at those coordinates. Allowing all four horizontal coordinates to vary within their rounding intervals lowers the infimum to about 1.07 m/s^2. Under our force laws, the horizontal component of non-drag acceleration toward the inside of that turn must reach at least 0.97 m/s^2 at some instant (Appendix C.2). These observations resolve a turn while leaving its cause and the spin rate uncertain.

In still air the horizontal path under drag alone must be a straight line through release. The slope intervals $[(y_i - \epsilon)/(x_i + \epsilon), (y_i + \epsilon)/(x_i - \epsilon)]$ first touch at $\epsilon = 7.279$ mm for this kick. A 5 mm per-axis bound resolves the turn, whereas a 1 cm bound admits a straight path.

2 Flight Models and Prediction Method

2.1 Aerodynamic Forces and Model Parameters

Write $\boldsymbol{v} = \dot{\boldsymbol{r}}$ for velocity over the ground, $\boldsymbol{u} = \boldsymbol{v} - \boldsymbol{w}$ for velocity relative to the air, and $U = \|\boldsymbol{u}\|$. The three force models give accelerations

$$\begin{aligned} \boldsymbol{a}_{\mathrm{D}} &= \boldsymbol{g} - \kappa U \boldsymbol{u}, \\ \boldsymbol{a}_{\mathrm{S}} &= \boldsymbol{a}_{\mathrm{D}} + \boldsymbol{\beta} \times \boldsymbol{u}, \\ \boldsymbol{a}_{\mathrm{E}} &= \boldsymbol{g} - K C_D U \boldsymbol{u} + K C_L U^2 \boldsymbol{n}. \end{aligned} \tag{3}$$

Model D has quadratic drag. S adds a transverse force with constant coefficient vector $\boldsymbol{\beta}$. In E, $K = \rho \pi R^2/(2m)$ and lift acts along the unit vector $\boldsymbol{n}$ in the direction of $\boldsymbol{\omega} \times \boldsymbol{u}$. Lift vanishes when the cross product is zero. We use $g = 9.81$ m/s^2 downward, radius $R = 0.11$ m, mass $m = 0.43$ kg and air density $\rho = 1.20$ kg/m^3. With inputs in SI units, the coefficients for E are

$$\begin{aligned} C_D &= 0.155 + \frac{0.346}{1 + \exp[(U_c - 12.19)/1.309]} + \delta, \qquad U_c = \min(29.9, \max(9.3, U)), \\ C_L &= 0.38 \tanh\left(5R\|\boldsymbol{\omega} \times \boldsymbol{u}\|/U^2\right). \end{aligned} \tag{4}$$

For the speed-dependent part of C_D, Goff and Carré [2] fitted their Eq. (3) and Fig. 1 to the nonspinning Adidas Teamgeist data of Asai et al. [3], whose tunnel tests of nonrotating balls covered 7–35 m/s. We clip the speed argument to 9.3–29.9 m/s using the initial-speed range of a separate 29-flight study [4]. The offset δ, the saturating lift law and the spin limits are modelling choices for the unspecified ball. In S, $\boldsymbol{\beta}$ represents an effective transverse-force coefficient.

We assign D, S and E prior probabilities $0.2, 0.4, 0.4$ and draw wind uniformly from a horizontal disk of radius 3 m/s. Launch-velocity components are independently uniform over $[18, 38]$, $[-12, 12]$ and $[1.5, 11]$ m/s along x, y, z. For D and S, κ is uniform on $[0.006, 0.024]$ m^{-1}. The components of $\boldsymbol{\beta}$ in S and $\boldsymbol{\omega}$ in E are uniform within ± 0.8 s^{-1} and ± 65 rad/s, respectively. In E, δ is uniform on $[-0.04, 0.27]$.

2.2 Force and Time Scales

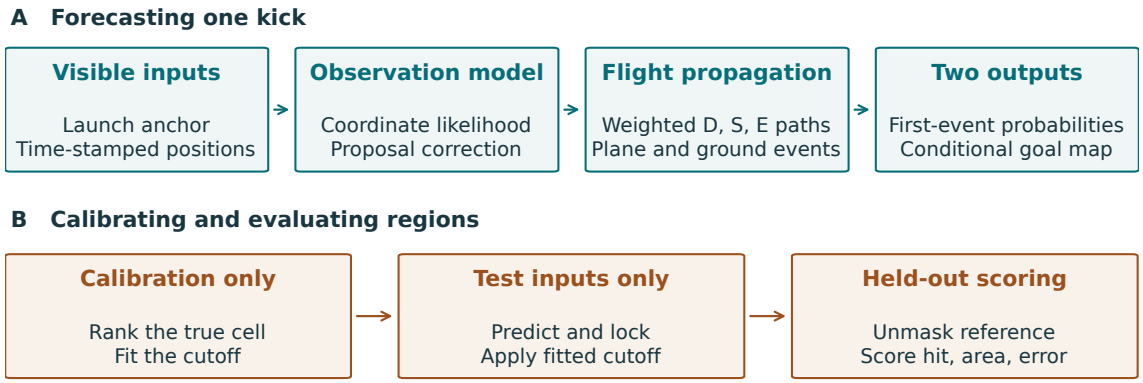


Figure 2. From early positions to a forecast. A follows one kick from its observed positions to first-event probabilities and a location map conditional on direct entry. B shows how we calibrate and test regions on separate kicks (Sec. 6). Only scoring uses the held-out reference crossings.

The first displacement gives a speed scale $U_0 \simeq 32$ m/s. At $\kappa \simeq 0.020$ m^{-1}, the 0.4 s flight lasts about 0.26 of the drag time $(\kappa U_0)^{-1} \simeq 1.56$ s. A transverse coefficient of 0.5 s^{-1} gives an initial acceleration scale of 16 m/s^2. A residual transverse acceleration of 1 m/s^2 sustained over the remaining 0.30 s would accumulate about 4.5 cm of displacement.

Constant wind also gives an energy check. Because $\boldsymbol{u} \cdot (\boldsymbol{\beta} \times \boldsymbol{u}) = 0$ and $\boldsymbol{u} \cdot \boldsymbol{n} = 0$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\frac{U^2}{2} = \boldsymbol{u} \cdot \boldsymbol{g} - d(U)U^2, \qquad d(U) \geq 0,$$

where $d(U)$ is the nonnegative drag rate. A transverse force turns the air-relative velocity, while drag dissipates air-relative kinetic energy. We use this identity to bound acceleration in the match records.

2.3 Trajectory Weighting and Goal-Plane Mapping

For each draw of wind and aerodynamic parameters, we estimate launch velocity from the latest visible position under gravity and correct the estimate three times by integrating to that position. We draw velocities from a local uniform distribution centred on this estimate or from the original prior with probability 2%. The local half-width is the position scale divided by the latest observation time, separately on each axis. The scale is 7 mm under rounding and 26 mm for the tracking likelihood in Z09 and Z10. Under rounding alone, a trajectory inside every observed interval receives the prior-to-proposal density ratio. All others receive zero weight. For parameters θ, proposal density q_M and likelihood L,

$$w_{Mj} = \frac{p(\theta_{Mj} \mid M)L(D \mid \theta_{Mj}, M)}{q_M(\theta_{Mj})}, \quad \widehat{Z}_M = \frac{1}{N_M}\sum_j w_{Mj}, \quad \widehat{p}(M \mid D) \propto p(M)\widehat{Z}_M. \tag{5}$$

Sampling nuisance parameters from their priors cancels those densities in the ratio. The correction for launch velocity remains. The mean weight $\widehat{Z}_M$ estimates the evidence for force model M. Systematic resampling limits propagation to 512 trajectories per model per test kick, or 8,192 per model per seed for the prescribed kick. Both prescribed-case runs stayed below their cap.

We use fourth-order Runge–Kutta steps of 25 ms through the two prescribed observations and at most 12.5 ms thereafter. Linear interpolation locates plane crossings and ground contact within a step. For the four flights in the original solver check, independent integrations including SciPy DOP853 [5] agreed within 0.171 mm in event position and 0.038 ms in time.

First-event probabilities sum the propagated weights. Direct-entry weights are normalized on a 30×10 equal-area goal grid. The modal cell centre gives the point. The raw region takes cells in decreasing order of mass to reach 90% (Fig. 2). Zero total weight leaves the forecast undefined. Zero direct-entry weight leaves only the location undefined. The effective sample size $\mathrm{ESS} = 1/\sum_j \bar{w}_j^2$ uses normalized weights including model and timing mixtures. We flag test ESS below 25. The prescribed case requires pooled ESS of at least 200 and ESS of at least 25 for each model with posterior mass of at least 10%. These are numerical checks on weight concentration.

3 Results for the Prescribed Kick

3.1 Most Probable Cell and Joint Prediction Region

Conditional on direct entry, the most probable cell contains 29.1% of the probability (Fig. 3). Its centre, approximately $(Y, Z) = (2.56, 2.07)$ m, lies about 1.10 m inside the positive-y post and 0.37 m below the crossbar. The six most probable cells form the rectangle

$$2.196 \le Y \le 2.928 \text{ m}, \qquad 1.708 \le Z \le 2.196 \text{ m}. \tag{6}$$

Together these cells cover 0.357 m^2 and contain 90.5% of the conditional probability. These figures summarize masses on the specified grid. The median crossing time from release is 0.399 s, with a central 90% interval of 0.392–0.407 s conditional on exact frame times. Section 5 examines uncertain frame timing.

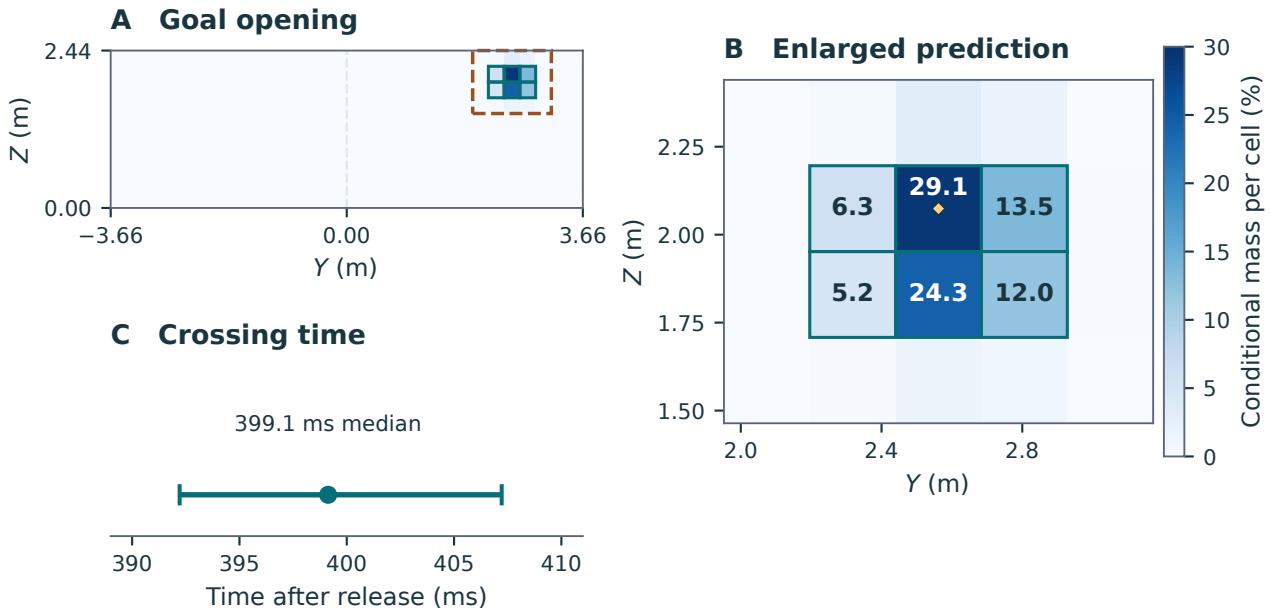


Figure 3. Predicted location and time of direct entry. Results use exact frame times, centimetre rounding and the nominal force-model mixture. A locates the enlargement. B shows probability per 0.244 m square cell with the same colour scale. The outline encloses the six most probable cells and the diamond marks the modal centre. C gives the 5th, 50th and 95th percentiles of the crossing time.

All 3,469 positive-weight trajectories from two independent runs of 131,072 proposals per model reached the opening, giving a finite-sample estimate $\widehat{P}(G) = 1$. A rare miss could remain unsampled. The model-stratified importance weights do not give a binomial confidence bound on the population probability. Posterior model weights were $0, 0.346, 0.654$ for D, S and E. The zero weight for D also follows from the wind bound in Eq. (2). The crossing of the prescribed kick has no measured reference.

3.2 Constraints on Physical Parameters

Replaying the same two runs gives the distributions in Table 2. In model E, the median release lift coefficient shifts from 0.277 under the prior to 0.220 given the readings. The spin-speed interval remains broad and the wind-speed distribution changes little. Only 6.3% of E's posterior weight puts any spin component in the outermost 2% at either face of its allowed interval, compared with 11.5%

under the prior. The fit gives little support for a spin rate near the imposed limit. An independent spin measurement would distinguish among the remaining explanations.

Table 2. Parameters of the prescribed kick. Each entry gives the median [5th, 95th percentile] within the specified force model. The last column compares posterior and prior mass in the top 2% of the scalar range. For C_L the permitted launch and spin ranges give a maximum of 0.37958 at release. For spin speed it is the corner of the component cube, $65\sqrt{3}$ rad/s. Zero denotes no sampled mass in the band.

Model	Quantity	Prior	After both readings	Upper band (%)
S	κ (m^{-1})	0.0150 [0.0069, 0.0231]	0.0198 [0.0149, 0.0233]	2.83 / 2.00
E	C_L (release)	0.277 [0.107, 0.356]	0.220 [0.134, 0.301]	0 / 0.38
E	Spin speed (rad/s)	64.0 [29.7, 89.8]	55.9 [29.9, 81.3]	0 / 0.004
S	Wind speed (m/s)	2.12 [0.67, 2.92]	2.03 [0.61, 2.90]	3.00 / 3.96
E	Wind speed (m/s)	2.12 [0.67, 2.92]	2.08 [0.66, 2.91]	3.31 / 3.96

3.3 Sensitivity of the Most Probable Cell

Both seeds selected the cell centred at $(2.562, 2.074)$ m. Varying E's share of the combined S/E prior from 10% to 90% also preserved this cell. A finer grid moved the modal centre to $(2.501, 2.013)$ m, and a half-cell shift of the original grid gave $(2.684, 1.952)$ m. The extra digits identify a cell on each grid. The six-cell region in Eq. (6) quantifies the nominal spatial uncertainty. The continuous conditional mean, $(2.603, 1.970)$ m, minimizes posterior squared distance. Our aerodynamic point forecasts use the centre of the cell with greatest probability at the stated resolution.

Gravity alone, with launch velocity fitted by least squares, predicts $(3.217, 1.957)$ m, just 0.44 m inside the near post. The aerodynamic modal cell lies 0.65 m farther toward the centre. This difference measures the combined effect of refitting the launch state and changing the force model and point estimator. A constant-acceleration fit through all three positions gives $(2.495, 1.920)$ m.

4 Limits of Two-Frame Prediction

4.1 Ambiguity at Exact Coordinates

How closely can any point forecast fit every admissible flight? In model S we fixed the wind at zero and β_x at either -0.5 or $+0.5$ s^{-1}. For each choice we fitted the six coordinates at 50 and 100 ms by varying the three launch-velocity components and κ, β_y, β_z. Both fits lie within the prior ranges, with maximum coordinate residual below 5×10^{-13} m. An independent integration with MATLAB `ode45` reproduced both crossings [6].

The two fitted drag coefficients agree to the displayed precision at 0.02007 m^{-1}. The fitted (β_y, β_z) pairs are $(-0.206, -0.463)$ and $(0.085, -0.251)$ s^{-1}, respectively. At any instant, a component of $\boldsymbol{\beta}$ along $\boldsymbol{u}$ contributes no transverse force. As gravity turns the velocity, that component can influence the later flight. Small adjustments in launch velocity allow different force histories to agree at both early frames.

The two flights yield identical centimetre readings but cross at $Q_A = (2.57378, 2.00251)$ m and $Q_B = (2.62831, 1.93396)$ m, 8.76 cm apart. Any point predictor must give them the same $\widehat{Q}$. The triangle inequality then gives

$$\max\{\|\widehat{Q} - Q_A\|, \|\widehat{Q} - Q_B\|\} \;\geq\; \tfrac{1}{2}\|Q_A - Q_B\| \;\approx\; 0.0438 \text{ m}. \tag{7}$$

This pair establishes nonuniqueness even when the displayed centres are fitted exactly. Its 4.38 cm floor is specific to these two flights. For comparison, mean two-frame errors on separate

synthetic entries were 17.0 cm under rounding and 25.5 cm with tracking errors (Table 5). These empirical averages measure a different quantity from worst-case error for the prescribed observations. Extending the fitted β_x values to the permitted endpoints $\pm 0.8\ \mathrm{s}^{-1}$ increases the crossing separation to 13.98 cm and raises the exact-centre floor to 6.99 cm.

4.2 Ambiguity under Centimetre Rounding

In a measurement system it is the *displayed coordinates* that have to match. We searched the convex hull of the 3,469 stored goal crossings for a widely separated pair. The two selected flights both use model S, with different admissible winds, launch velocities and transverse-force coefficients. Independent adaptive integrations put every early coordinate strictly inside its required centimetre bin, with the closest coordinate 0.043 mm from a bin edge. Both flights reached the open goal before the ground. Their crossing points were $(2.290, 2.329)$ and $(2.906, 1.644)$ m, separated by 0.922 m. By Eq. (7), **any single prediction from these displayed readings must miss at least one of the pair by 0.46 m**. This is an exhibited worst case. The sample search gives no upper bound on the full feasible set or probability that either flight occurs.

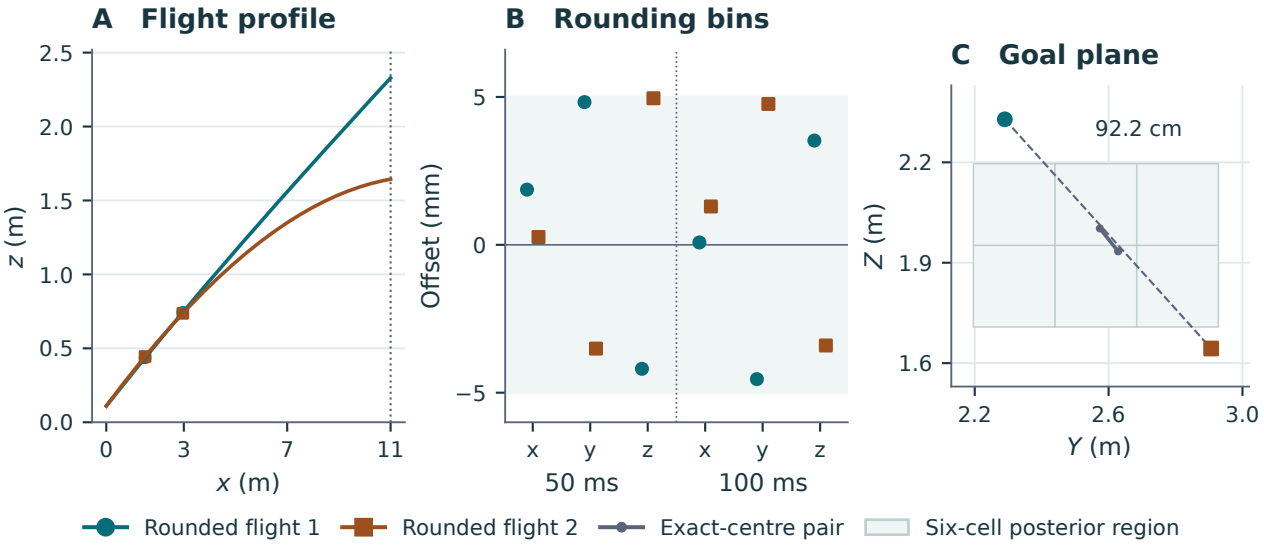


Figure 4. Identical readings can lead to different crossings. A shows two admissible model-S flights in side view. B gives coordinate offsets from the displayed values, within every centimetre bin. C shows their crossings and those of the exact-centre pair on equal spatial scales. Half the 92.2 cm separation gives the rounded pair's worst-case point-error lower bound.

5 Measurement Uncertainty

5.1 Position, Timing, and Rounding Errors

For axis a, let d_{ia} denote the position in frame i displayed to the nearest $q = 0.01$ m. We model the displayed value as

$$d_{ia} = \mathcal{R}_q[r_a(t_i + \tau; \theta, M) + b_a + e_{ia}], \quad b_a \sim N(0, \sigma_b^2), \quad e_{ia} \sim N(0, \sigma_e^2). \tag{8}$$

Here b_a is a bias common to the frames of one kick, e_{ia} is independent across frames and axes, and τ shifts all the frame times together. Conditional on b, τ, the probability of the displayed readings is

$$L = \prod_{i,a} \left[\Phi\left(\frac{d_{ia} + q/2 - r_a(t_i + \tau) - b_a}{\sigma_e} \right) - \Phi\left(\frac{d_{ia} - q/2 - r_a(t_i + \tau) - b_a}{\sigma_e} \right) \right]. \tag{9}$$

Here Φ is the standard normal cumulative distribution function. With rounding alone, L is the indicator that every coordinate lies in its interval. We sample b from its prior and integrate a shared uniform timing offset by Gauss–Legendre quadrature. The synthetic tests use five or seven nodes. We check the prescribed-kick timing calculation separately below.

A position error small enough to miss by eye can dominate the inferred acceleration. For an exact release coordinate and frame errors on one axis with common variance σ^2 and correlation ρ_e, the acceleration error inferred from the second difference at spacing h has variance

$$\operatorname{Var}\left(\frac{e_2 - 2e_1}{h^2}\right) = \frac{(5 - 4\rho_e)\sigma^2}{h^4}. \tag{10}$$

A common timing shift contributes $-v\tau/h^2$ to the acceleration inferred from uniform motion. At $v = 30$ m/s and $h = 50$ ms, a 1 ms shift produces an apparent acceleration of 12 m/s^2. A spatial bias cancels from the second difference if it affects release and both frames equally. If the release coordinate is exact, a bias shared only by the two frames survives and can resemble aerodynamic curvature.

For the prescribed readings, we allowed a shared phase τ uniform on $[-1, +1]$ ms and refitted launch velocity at each possible pair of true frame times. Weighting the phases by the evidence from both readings gave a median release-to-crossing time of 394 ms and a central 90% interval of 383–406 ms on direct entries. Measured from the *actual second frame*, the corresponding median was 294 ms and the interval was 282–306 ms. The calculation used 24 quadrature nodes and 524,288 proposals per force model per seed, with pooled ESS above 7,600. At 12 nodes the three quantiles of each distribution changed by less than 1 ms. These intervals are conditional on the assumed timing prior.

5.2 Sensitivity to Measurement Assumptions

We compared the nominal case with fourteen one-at-a-time variants and refitted the same readings, using the nominal two seeds and 131,072 proposals per model per seed. Table 3 and Fig. 5 use displacement of the continuous conditional mean to track movement independently of the grid. Region areas use the original grid.

Table 3. Nominal forecast and nine selected variants with adequate sampling support. Shift compares the continuous means conditional on entry. Area is the raw 90% conditional region. The final column gives the weighted fraction of trajectories entering directly, before conditioning. Fig. 5 includes all fourteen variants.

Changed assumption	Shift (cm)	Area (m^2)	$\widehat{P}(G)$ (%)
Nominal assumptions	0.00	0.357	100.0
Coordinate half-width: 10 mm	14.11	0.893	95.6
Both observation times: +1 ms	3.73	0.357	100.0
Wind cap: 0 m/s	0.83	0.357	100.0
S: halve β component limits	3.97	0.357	100.0
E: halve ω component limits	9.08	0.357	100.0
E: ball mass 410 g	0.25	0.357	100.0
E: ball mass 450 g	0.45	0.357	100.0
S as the sole force model	0.67	0.417	100.0
6 mm independent + 8 mm shared noise	23.70	1.072	93.4

For this kick the largest well-supported change came from the measurement assumptions. Doubling the coordinate bound moved the conditional mean by 14.1 cm and enlarged the six-cell region to 15 cells. With tracking error it moved 23.7 cm, and the region grew to 18 cells. The estimated

direct-entry probabilities also fell to 95.6% and 93.4%. A fixed +1 ms shift in both observation times moved the mean by 3.73 cm while retaining six cells. A tighter 2.5 mm bound and a −1 ms shift gave ESS of only 68 and 1. Neither 5 ms shift yielded a positive-weight sample.

Five changes to wind, spin ranges and ball mass moved the mean by 0.25–9.08 cm while retaining six cells. Equal area can hide a different hotspot: halving E's spin range moved the modal centre one 0.244 m cell across the goal. Using S alone enlarged the region to seven cells. The 410–450 g mass limits come from the match-ball specification [8]. Early coordinate error is amplified on extrapolation (Appendix A), while the force laws restrict admissible curvature. The ranking here depends on this kick, the grid and the tested ranges.

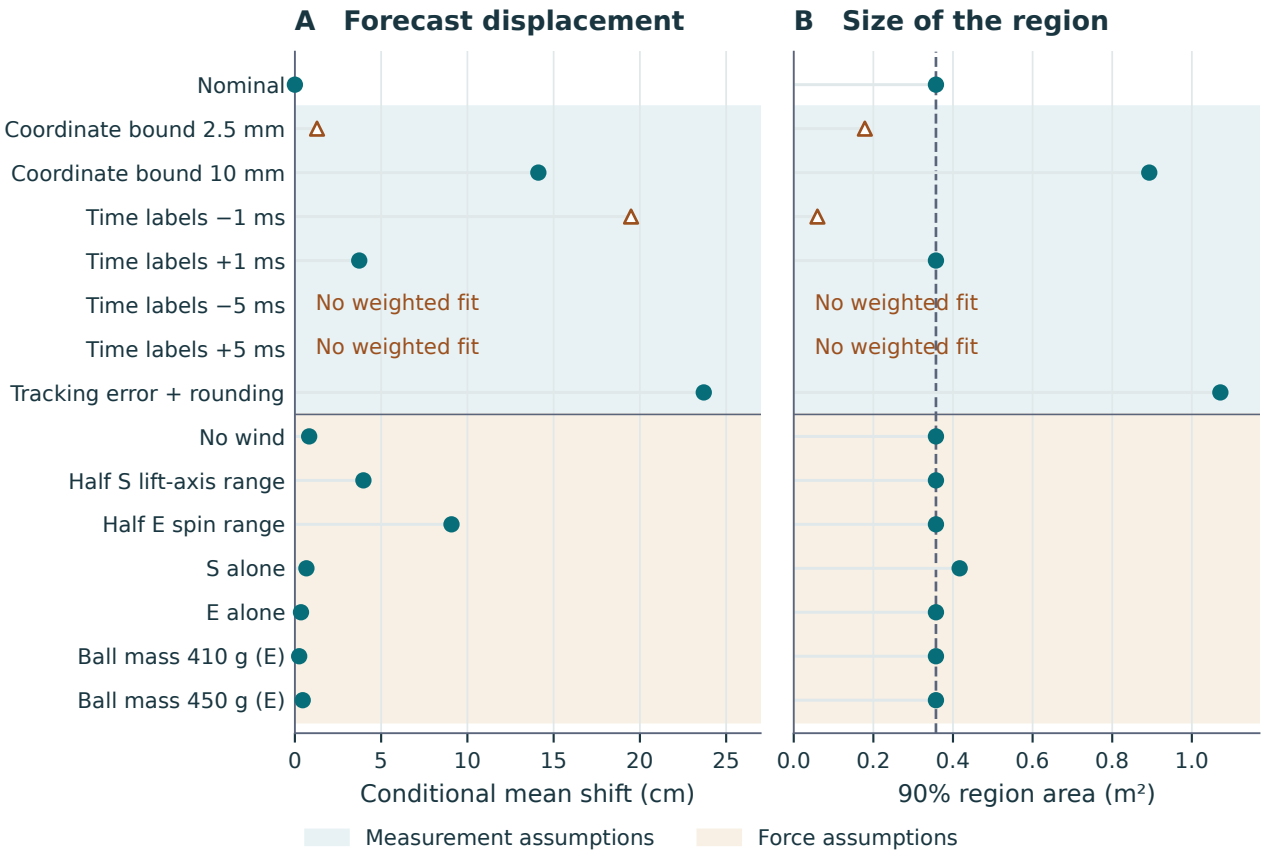


Figure 5. Sensitivity of the mean crossing point and prediction region area. A shows displacement of the conditional mean. B shows the raw 90% region area and its nominal value (dashed). Blue shading groups measurement changes and ochre shading groups force assumptions. Open triangles mark ESS below the prescribed-case threshold of 200. Rows with zero positive weight have no estimate.

5.3 Combined Timing and Tracking Errors

We refitted the prescribed readings in a 2×2 comparison. The two time settings fix the frame times or marginalize a shared uniform offset over ± 1 ms. The two position settings use centimetre rounding alone or add independent 6 mm noise and a shared 8 mm bias per coordinate before rounding. We retained the same force priors and 131,072 proposals per model per seed. The phase runs divided this budget among 12 quadrature nodes. For the noisy likelihood we widened the velocity proposal to $0.025\,\mathrm{m}/t$ with the corresponding importance correction. These are forecasts at the second recorded frame for the same displayed inputs.

Adding the tracking errors enlarged the region from six to 18 cells in both time settings. The estimated direct-entry probability fell to about 93%, and marginalizing the clock changed the median crossing time. The region cell counts are stable across these two clock settings, while their shapes can differ. This comparison uses one prescribed kick and one pair of error scales. The two seed estimates differ by up to 2.1 cm in conditional map centroid, so changes on that scale should be read with the sampling variability in mind.

Table 4. Joint observation assumptions for the prescribed kick. All rows pass the prescribed-case sampling checks. The cell count and area describe the raw conditional region. Arrival is conditional on direct entry and measured from release.

Positions	Clock	Cells	Area (m^2)	Entry (%)	Median (ms)
Rounding	Exact	6	0.357	100.0	399.1
Rounding	Uniform ±1 ms	6	0.357	100.0	394.5
Tracking + rounding	Exact	18	1.072	93.4	397.7
Tracking + rounding	Uniform ±1 ms	18	1.072	93.3	396.3

6 Forecast Validation at 100 ms

6.1 Paired Forecasts at a Common Deadline

At the common 100 ms input cutoff, one aerodynamic predictor also receives the 50 ms reading. Both use the same priors, force laws and solver, with 8,192 proposals and up to 512 propagated trajectories per model. Each predicts its modal cell centre. The gravity control uses both readings to compute a continuous crossing and a binary direct-entry probability.

Z09 generated 90 calibration kicks and 120 test kicks with fresh seeds. A 1 ms reference solver used different force laws and balanced blocks of zero-spin, stable-spin and low-spin wake flights (Appendix B.1). We calibrated regions on 55 direct entries. We scored locations on the 71 test entries and events on all 120 kicks.

We rounded each kick's exact coordinates to the nearest centimetre. For a second set of readings, we first added independent Gaussian errors in each frame (6 mm standard deviation per coordinate) and a Gaussian bias shared by both frames (8 mm standard deviation per coordinate). Release position and frame times were exact in both sets. For noisy readings the predictor used the same error model as the generator. Appendix A treats timing errors separately.

We prespecified the methods, 50–100% coverage targets (90% for the primary comparison), and subgroups by spin and distance to the goal edge. Methods and profiles share one inference seed per kick. Predictions used only inputs and were hashed before reference scoring.

6.2 Calibration of Prediction Regions

For each method and measurement condition, calibration sets a threshold for including cells. In the aerodynamic forecasts, rank the grid cells $j_1, \dots, j_J$ by decreasing probability $H_D(j)$ conditional on direct entry, breaking ties in a fixed order. The score of an occupied cell is the mass of the cells ahead of it,

$$S_D(j_k) = \sum_{\ell<k} H_D(j_\ell). \tag{11}$$

An empty cell scores one. For gravity, a cell whose centre lies distance d_j from the ballistic crossing scores $d_j/(L + d_j)$, where L is the goal diagonal. Thresholding this score amounts to choosing a

radius about the crossing.

For a target coverage c and n calibration entries, sort the scores of their true cells and take the $\lceil (n+1)c \rceil$-th smallest as q_c. Include all cells with $S_D(j) \le q_c$. If the rank exceeds n, set $q_c = 1$. Z09 assigns score one to all cells for an unavailable prediction, so $q_c = 1$ yields the whole goal even in this case. Z10 counts unavailable regions as misses of zero area. All reported direct entries in these tests had usable forecasts for the main methods. Split calibration gives marginal coverage for exchangeable direct entries [9]. Here the balanced family blocks define a synthetic test population with empirical hit rates. Paired bootstrap intervals resample test kicks with the fitted cutoffs fixed.

Table 5. Forecasts at the 90% calibration target. Hits, mean area and mean point error use the 71 direct entries. The Brier score for direct entry uses all 120 test kicks. The gravity control uses both readings and a calibrated radius.

Profile	Forecast	Hits	Area (m^2)	Mean error (m)	Brier
rounded	100 ms only	64/71	2.003	0.208	0.070
rounded	50 + 100 ms	67/71	0.381	0.170	0.018
rounded	Gravity + radius	70/71	1.222	0.186	0.083
tracking	100 ms only	62/71	1.825	0.216	0.075
tracking	50 + 100 ms	60/71	0.859	0.255	0.078
tracking	Gravity + radius	69/71	1.255	0.199	0.075

6.3 Coverage under Centimetre Rounding

At the 90% target, two frames covered 67/71 entries in 0.381 m^2 on average, compared with 64/71 in 2.003 m^2 using the late frame alone (Table 5). The respective 95% Wilson intervals for coverage were 86.4–97.8% and 81.0–95.1%. At their 95% targets, both methods covered 70/71 entries. The two-frame region averaged 0.585 m^2, compared with 3.482 m^2 for the late frame. Gravity covered 70/71 in 1.222 m^2 at its 90% target (Fig. 6). Equal counts can include different kicks.

Adding the early frame lowered point error on 13 entries, left it unchanged on 54 and raised it on four (Fig. 7). The mean reduction was 3.80 cm (95% interval 0.22–7.62 cm from 4,000 paired resamples of kicks). The mean reduction relative to gravity had a 95% interval of −2.29 to 5.85 cm. The Brier score for direct entry was 0.018 with both frames, 0.070 with the late frame and 0.083 for the binary gravity forecast.

6.4 Coverage under Tracking Noise

At the same 90% target, the two-frame region was the smallest but covered only 60/71 noisy crossings, compared with 62/71 for the late frame and 69/71 for gravity. All eleven misses by the two-frame forecast came from the stable-spin group (12/23 hits). It covered 24/24 in each of the zero-spin and wake groups. Gravity covered 21/23 stable-spin entries (Table 13).

At 67/71 observed hits, gravity used 0.933 m^2 at its 80% target and two-frame aerodynamics used 1.385 m^2 at its 95% target. These two targets yielded equal coverage on this particular test sample. For the 71 direct entries, the aerodynamic grid centre had a mean point error 5.57 cm larger than the continuous gravity forecast (paired 95% interval 1.17–10.25 cm). The intervals for both the point-error and Brier-score differences between two-frame and late-frame aerodynamics included zero.

Estimating acceleration from the readings divides the second position difference by h^2, with $h = 50$ ms. A bias shared by the two frames survives in that difference when the release position is



Figure 6. Mean region area at equal observed hit counts. A and B plot hits against mean area on a log scale for the same 71 entries. Open markers denote the primary 90% calibration target. In C and D, targets from the prespecified grid were matched after scoring. C uses 95% for the two aerodynamic methods and 90% for gravity. D uses 95% for two-frame aerodynamics and 80% for gravity. No late-frame point matched 67/71.

fixed independently (Eq. (10)). Gravity fixes the acceleration. Its relative precision here is consistent with the difficulty of estimating transverse curvature from noisy readings, even when the noise model is known.

6.5 Forecast Availability and Runtime

Each method produced a point for all 71 direct entries and an event probability for all 120 kicks under both measurement conditions. Only one run, a noisy late-frame forecast, fell below 25 effective samples. Even with a defined point, the noisy two-frame region filled the entire goal at its 97.5% target. Appendix B.4 gives the near-edge counts, full target comparisons and runtime distributions.

In the local Python run, two-frame inference and goal-map construction took 84 ms at the median and 94 ms at the 95th percentile with rounding alone. With tracking errors the times were 121 and 131 ms. At the test sampling budget, 131 ms is about 44% of the nominal 299 ms remaining after the last reading. Capture, reconstruction, transfer and goalkeeper movement require additional time, and the crossing time varies by kick.

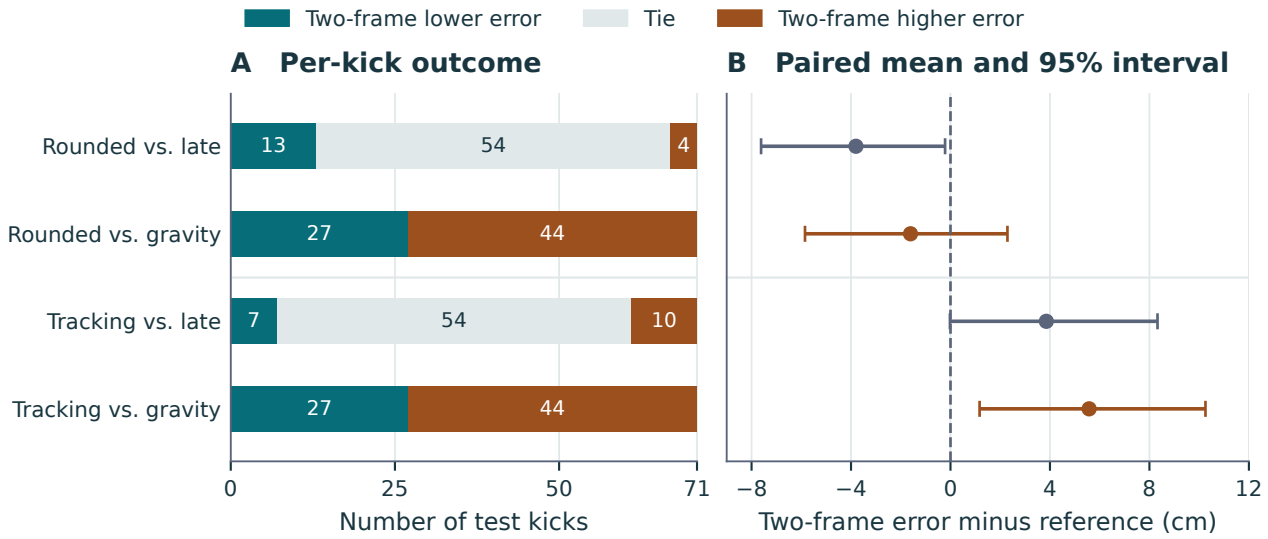


Figure 7. Point errors on the same 71 direct-entry kicks. A counts kicks on which two-frame aerodynamics has a smaller, equal or larger error than each comparison method. B shows the mean paired difference and its 95% shot-bootstrap interval. Negative values favour two-frame aerodynamics.

6.6 Independent Stable-Spin Test

The 12/23 aerodynamic hits on noisy stable-spin entries in Z09 prompted a second test (Z10). We drew 300 new calibration kicks and 600 new test kicks, balanced by the same three spin families, and used DOP853 to generate the reference positions and first events. Each predictor saw only the 50 and 100 ms readings. We locked the forecasts, fitted a 90% threshold for each method on the 180 calibration entries pooled across families (rank 163), then scored the held-out crossings.

With tracking errors, two-frame aerodynamics covered 343/368 direct entries overall (93.2%) and 90/110 with stable spin (81.8%). The stable-spin group accounted for 20 of the 25 misses (Fig. 8A). Gravity covered 77/110 stable-spin entries. Aerodynamics alone covered 31 and gravity alone covered 18, a paired difference of +11.8 percentage points (95% interval 0.0 to +23.6). The *post hoc* threshold exchange in Table 6 keeps each map and gravity forecast fixed. On Z10, switching from the Z09 cutoffs to the Z10 cutoffs expands the mean aerodynamic region from 0.875 to 1.019 m^2 and shrinks gravity's from 1.148 to 0.845 m^2. The coverage ranking reverses as these operating points change.

Table 6. Noisy stable-spin entries. Each row applies both methods' previously fitted cutoffs from the named batch to the same test entries. This threshold exchange reuses the two synthetic tests.

Test	Cutoffs from	Hits		Mean area (m^2)	
		Aero	Gravity	Aero	Gravity
Z09	Z09	12/23	21/23	0.888	1.261
Z09	Z10	14/23	19/23	1.033	0.922
Z10	Z09	78/110	94/110	0.875	1.148
Z10	Z10	90/110	77/110	1.019	0.845

Appendix B.5 quantifies how frame noise propagates to the inferred acceleration. Cleaner measurements improved the point forecast. In the stable-spin group, rounding alone lowered mean point error from 0.421 to 0.258 m. In the low-spin wake group, however, the smaller regions under rounding covered fewer entries (Table 7). Both profiles use the same 90% target with separately fitted cutoffs, so their point accuracy and regional coverage can move in different directions.

Keeping the noisy observations and halving the prior ranges for S and E gave a stable-spin mean

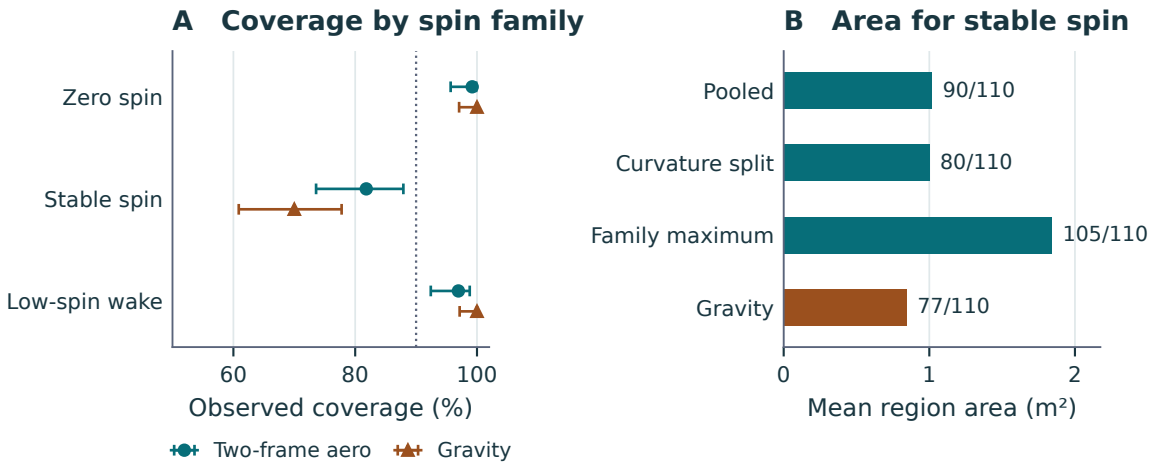


Figure 8. Two-frame regions under tracking noise in Z10. A shows coverage and 95% Wilson intervals for 127 zero-spin, 110 stable-spin and 131 low-spin wake entries. Each method uses its own pooled cutoff. The dotted line marks the 90% calibration target. B gives mean areas and hits on the 110 stable-spin entries. The first three bars apply a pooled cutoff, a two-frame curvature split at twice the per-axis noise scale (Appendix B.5.3), or the largest family cutoff. Gravity uses its own pooled radius.

Table 7. Two-frame aerodynamics on the same Z10 entries under two observation profiles. Each profile uses a matching likelihood and its own pooled 90% cutoff. No-spin hits are 126/127 under both profiles. Area and error average all 368 entries.

Profile	All hits	Stable spin	Low-spin wake	Area (m^2)	Error (m)
Rounding	336/368	100/110	110/131	0.357	0.174
Tracking + rounding	343/368	90/110	127/131	1.003	0.257

error of 0.379 m, with 94/110 hits in 0.579 m^2 after calibration. Applying the largest family cutoff to the original forecasts covered 105/110 stable-spin entries in 1.843 m^2. That rule uses family labels in calibration and the same cutoff for every test kick. It changes the region while retaining each original point and event probability.

6.7 Point and Event Forecast Scores

We reanalysed the frozen Z09 and Z10 tests to check sensitivity to how each method summarizes the forecast. For points, we replaced the aerodynamic modal cell by the probability-weighted centroid of the same grid. This minimizes expected squared distance for the discrete map. We score both methods by mean Euclidean distance to the observed crossing. The deterministic ballistic point remains its continuous crossing. For events, we used the signed margin

$$m_b = \min(3.66 - |Y_b|,\ Z_b - R,\ 2.44 - Z_b)$$

from the gravity-only fit, extrapolated to the plane. All fitted downfield velocities were positive in these batches. Under gravity alone, the height is concave in time and the ball starts at R. Thus $m_b > 0$ is the ballistic condition for direct entry away from equality. We fitted a monotone logistic probability against the standardized margin using only each study's calibration kicks, with a unit ridge penalty on the slope. The fitted probability changes the event score only.

The centroid lowers mean point error relative to gravity under rounding and raises it under tracking errors in both batches (Table 8). The logistic probability improves gravity's Brier score over

Table 8. Retrospective comparison on locked synthetic tests. Point error averages the 71 Z09 or 368 Z10 direct entries. Brier scores use all 120 or 600 kicks, respectively. Each logistic fit uses only its study and profile's calibration data.

		Mean Euclidean error (m)		Event Brier score		
Test	Profile	Aero centroid	Gravity	Aero	Gravity binary	Gravity logistic
Z09	Rounding	0.144	0.186	0.018	0.083	0.051
Z09	Tracking	0.322	0.199	0.078	0.075	0.052
Z10	Rounding	0.157	0.198	0.033	0.087	0.066
Z10	Tracking	0.303	0.211	0.064	0.093	0.067

its binary prediction in every row. Aerodynamics still scores better under rounding. With tracking errors, the logistic forecast scores better on Z09. On Z10 the difference (aerodynamics minus logistic gravity) has a paired 95% test-kick bootstrap interval of $[-0.0157, +0.0090]$, with calibration held fixed. These controls reuse the existing test kicks. The prescribed-kick answer uses the highest-probability cell.

6.8 Lift Priors and Calibration Variability

In Z11 we halve the component half-width of S's effective transverse-force coefficient vector, E's spin vector, or both. The other force priors and the prior weights of the force models stay fixed. Conditioning can change the posterior model weights. We reuse random streams within each kick and draw 8,192 proposals per force model in every arm. The comparison reuses the 600 Z10 tracking-noise inputs at the 100 ms cutoff. Replayed forecasts for the original mixture, joint reduction and gravity agree with their archived counterparts. This isolates changes to the prior specification on a previously examined test cohort.

Table 9. Separate prior reductions on the same 110 stable-spin direct entries. Error is mean Euclidean distance from the modal point to the crossing. Rank one is best. All four hit counts use the original Z10 nominal aerodynamic cutoff of 0.8594. A common cutoff can admit different areas.

Prior reduced	Mean point error (m)	Mean true-cell rank	Fixed-cutoff hits
Nominal	0.421	11.92	90/110
S only	0.412	8.64	95/110
E only	0.406	7.11	101/110
S and E	0.379	6.31	99/110

Reducing S alone lowers mean point error by 0.94 cm and reducing E alone lowers it by 1.46 cm, with paired 95% test-kick intervals spanning zero in both cases (Fig. 9A). Reducing both lowers mean error by 4.17 cm (interval 2.08–6.28 cm). At the original cutoff, reducing E alone gives 101/110 hits, compared with 95/110 for S alone and 99/110 for the joint reduction (Table 9). The point and region comparisons therefore distinguish different effects of the priors.

We also generate eight independent calibration batches of 90 kicks, with 30 kicks from each reference force family per batch and fresh trajectory, observation and inference seeds. Each batch contains 47–56 direct entries. For each batch and method, we fit the pooled 90% cutoff and apply it to the same frozen Z10 test forecasts. Gravity lacked one region on a direct entry in each of calibration batches 3 and 8. These receive score one in the rank calculation. All test direct entries had usable regions. On their respective score scales, nominal aerodynamics has cutoffs from 0.735 to 0.904 and gravity from 0.065 to 0.083.

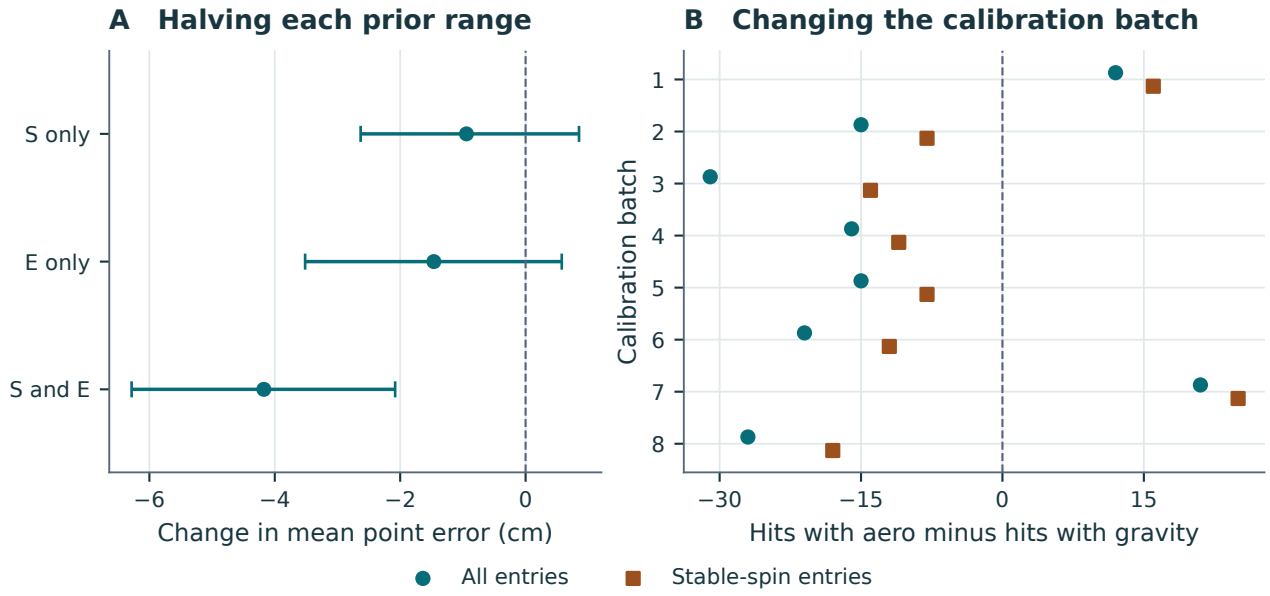


Figure 9. Sensitivity to lift priors and calibration data. A shows paired point-error changes from the nominal aerodynamic prior on 110 stable-spin entries, with 95% kick-bootstrap intervals. B subtracts gravity's hit count from nominal aerodynamics for each independent calibration batch, separately for all 368 direct entries and the 110 stable-spin entries. Both panels reuse the same Z10 test cohort.

On stable-spin entries, nominal aerodynamics covers 61–97 of 110 and gravity covers 72–101. The aerodynamic method has more hits in two batches and fewer in six, both on stable-spin entries and across all direct entries. In each of the eight batches and both groups, the method with more hits uses a larger mean region. For instance, calibration batch 1 gives 93 versus 77 stable-spin hits in 1.111 versus 0.885 m^2, while batch 3 gives 61 versus 75 in 0.638 versus 0.811 m^2. These comparisons quantify sensitivity to calibration within the declared synthetic population. The eight comparisons share test kicks, so they describe the spread across calibration batches.

7 Discussion and Limitations

7.1 Field Evidence and Model Limitations

The three 25 Hz match records yielded zero eligible cases for field validation because release timing could not be verified. In a retrospective check anchored to the first live-ball frame, none of the three pairs of readings was compatible with the nominal launch model. Allowing an in-flight anchor recovered a forecast for one kick. Appendix E documents the three records; Appendix E.4 specifies the measurements needed for a field test.

The forecast describes an uninterrupted flight under specified aerodynamic priors. Ground rebounds, interventions and collisions with the posts or crossbar could change the outcome near the goal edges, and the mix of forces in actual matches may differ from the synthetic population. The grid, measurement model and decision time therefore belong alongside any quoted point and region.

7.2 Goalkeeper Decision Time

With an exact frame clock the prescribed kick leaves a median of 299 ms for computation and movement after the last reading. Under the shared ±1 ms phase prior, the corresponding estimate is 294 ms (central 90%, 282–306 ms). In a separate synthetic test, shifting the two observations from 50/100 to 100/150 ms reduced mean point error from 0.182 to 0.126 m and left 50 ms less time to

act. Keeping all three frames at the later deadline gave more low-ESS forecasts (Appendix D). To translate either forecast into a save probability, one would also need the goalkeeper's initial position, movement onset and reach.

8 Conclusion

For the prescribed kick, the measured heading turns toward the centre relative to its first 50 ms chord. Under the stated wind and rounding limits the turn requires a transverse aerodynamic force. The modal cell is centred at $(2.56, 2.07)$ m, and six cells cover 0.357 m^2 and contain 90.5% of the conditional entry probability. Two force histories that generate the same displayed readings cross 0.92 m apart, illustrating how a concentrated probability map can coexist with a large worst-case error.

For the prescribed inputs, the tested tracking errors tripled the region area at either timing setting. At the common 100 ms cutoff, a second rounded reading sharply reduced the area at 70/71 observed hits. With tracking errors, gravity used less area at 67/71 observed hits in Z09, and its smaller mean point error persisted under the map-centroid comparison. The eight fresh calibration batches changed the coverage ranking on fixed test kicks. The method gaining hits also used more area. Reducing both lift priors lowered stable-spin point error on the Z10 cohort, while the separate point comparisons remained inconclusive. Independent release, early position and crossing measurements are needed to evaluate these choices on real penalties.

Appendix A Error Propagation at the Goal Plane

A.1 From Position Error to Crossing Error

A position error can change both where and when the ball crosses the goal plane. We can separate these effects in the constant-acceleration fit. For a fixed release position and readings at h and $2h$, its acceleration and launch velocity are

$$\widehat{\boldsymbol{a}} = \frac{\boldsymbol{r}_2 - 2\boldsymbol{r}_1 + \boldsymbol{r}_0}{h^2}, \qquad \widehat{\boldsymbol{v}}_0 = \frac{\boldsymbol{r}_1 - \boldsymbol{r}_0}{h} - \frac{h}{2}\widehat{\boldsymbol{a}}. \tag{12}$$

The prescribed readings give $\boldsymbol{v}_0 = (30.4, 9.4, 6.9)$ m/s and $\boldsymbol{a} = (-16, -16, -12)$ m/s^2. Extrapolate with this acceleration and write $X = 11 - x_0$, $D = v_{0x}^2 + 2a_x X$. For a first forward crossing with $D > 0$, the positive root has the numerically stable form

$$T = \frac{2X}{v_{0x} + \sqrt{D}}, \quad \dot{x}(T) = \sqrt{D}. \tag{13}$$

The control crosses at $T = 0.4050$ s and $(Y, Z) = (2.495, 1.920)$ m. The aerodynamic forecast appears in Eq. (6).

Let $\delta\boldsymbol{r}(T)$ be the change in position at the original crossing time. Requiring the perturbed trajectory to satisfy $x = 11$ m at its own crossing gives

$$\delta T = -\frac{\delta x(T)}{\dot{x}(T)}, \qquad \delta Q = P\,\delta\boldsymbol{r}(T), \qquad P = \begin{pmatrix} -\dot{y}(T)/\dot{x}(T) & 1 & 0 \\ -\dot{z}(T)/\dot{x}(T) & 0 & 1 \end{pmatrix}. \tag{14}$$

The local slopes convert longitudinal error into a shift across the goal. An error parallel to the velocity changes only the arrival time to first order, since $P\boldsymbol{v}(T) = 0$. The slopes can diverge as $\dot{x}(T)$ approaches zero. If the forward crossing disappears, its location derivative is undefined.

The Lagrange weights of the two readings are $\lambda_1 = s(2-s)$ and $\lambda_2 = s(s-1)/2$, where $s = T/h$. Let $\boldsymbol{e}_i$ be the spatial error in reading i. A common time offset $\tau > 0$ means that we fit the readings at their labelled times $h, 2h$ although both were taken τ later. To first order, the combined shift is

$$\delta Q = P\left[\lambda_1\boldsymbol{e}_1 + \lambda_2\boldsymbol{e}_2 + (\lambda_1\boldsymbol{v}(h) + \lambda_2\boldsymbol{v}(2h))\tau\right]. \tag{15}$$

At the nominal crossing the weights are -49.41 and 28.76. Allowing independent errors of up to 5 mm in each coordinate gives linear bounds of 0.439 m in Y and 0.424 m in Z. A common $+1$ ms offset shifts (Y, Z) by $(-0.123, -0.093)$ m to first order. Central differences with a $1\,\mu$s step reproduce the timing derivative to within 2×10^{-5} m/s. The large weights arise from extending an unconstrained quadratic well beyond the observations. The aerodynamic predictor restricts curvature through its force and launch bounds.

With the same positions reassigned to times $h + \tau, 2h + \tau$ and the quadratic refitted, $dT/d\tau = -26.52$ at zero. The exact forward roots for $\tau = -1, 0, +1$ ms are respectively 439.2, 405.0 and 383.1 ms. Holding the time labels fixed and instead generating later positions, as in Eq. (15), reverses the local derivative. This timing sensitivity belongs to the unconstrained quadratic. The aerodynamic timing interval in Sec. 5 weights physically admissible flights.

A.2 Covariance near a Smooth Crossing

Let Σ_e be the covariance of noise drawn independently in each frame, Σ_b that of a bias shared by the frames, and σ_τ^2 the timing variance. Assume independence among these sources. With $w = \lambda_1\boldsymbol{v}(h) + \lambda_2\boldsymbol{v}(2h)$, Eq. (15) gives

$$\mathrm{Cov}(\delta Q) = P\left[(\lambda_1^2 + \lambda_2^2)\Sigma_e + (\lambda_1 + \lambda_2)^2\Sigma_b + \sigma_\tau^2 w w^\top\right]P^\top. \tag{16}$$

Timing uncertainty acts along a single direction, Pw. The bias coefficient changes if the release shares the frames' translation. This covariance describes measurement uncertainty near a smooth crossing. Model discrepancy and sampling error contribute separately.

A.3 Joint Perturbations and Crossing Failure

We combined both timing limits and zero offset with every corner of the six-coordinate error box, counting a zero bound once. For each combination we refitted the quadratic and solved for its first forward plane crossing, ignoring the ground and goal edges. The largest displacement among sampled crossings need not bound the full box, whose interior may contain larger shifts.

Table 10. Joint perturbations of the quadratic control. The two linear bounds sum the absolute derivative contributions in each coordinate. The sampled shift is the largest Euclidean displacement among forward crossings. The last column counts trials with no forward root among all tested combinations.

Phase bound (ms)	Spatial bound (mm)	First-order Y (m)	First-order Z (m)	Sampled shift (m)	No root (count)
0	0	0.000	0.000	0.000	0/1
0	2.5	0.219	0.212	0.319	0/64
0	5	0.439	0.424	0.671	0/64
0	10	0.877	0.848	1.502	0/64
1	0	0.123	0.093	0.197	0/3
1	2.5	0.342	0.305	0.533	0/192
1	5	0.562	0.517	0.930	0/192
1	10	1.000	0.942	2.025	0/192
3	0	0.370	0.280	0.297	1/3
3	2.5	0.589	0.492	0.537	64/192
3	5	0.808	0.704	2.298	48/192
3	10	1.247	1.128	4.487	32/192

With bounds of 1 ms and 5 mm, all 192 combinations crossed the plane. The largest sampled displacement was 0.930 m, while the separate linear bounds in Y and Z were 0.562 and 0.517 m (Table 10). At 3 ms and 5 mm, 48/192 combinations lost the forward root. Widening a box changes the corners sampled, so the failure fraction can rise or fall. Propagating each perturbation to its own crossing captures both large shifts and the loss of the event.

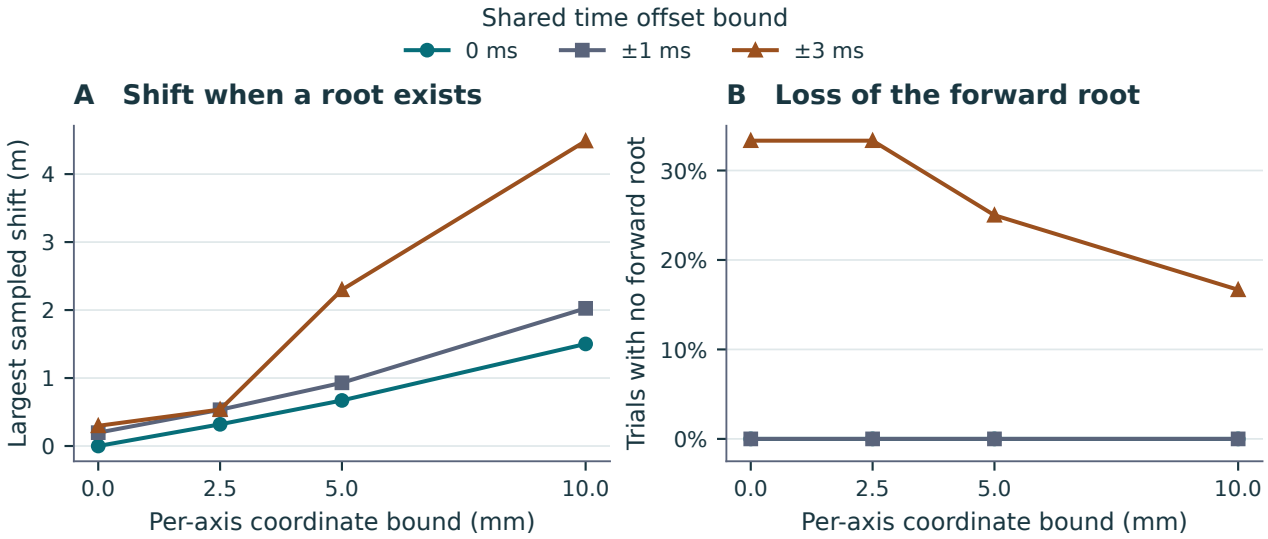


Figure 10. A perturbation can shift or remove the plane crossing. A gives the largest displacement among sampled quadratic fits with a forward root. B gives the fraction of tested combinations with no root. Each timing and coordinate combination is a deterministic trial.

Appendix B Synthetic Experiments and Forecast Robustness

The reference flights below are synthetic. We first define how they were generated, then separate three comparisons: early tests of sampling budget and measurement noise, the Z09 comparison at a common deadline, and the independent Z10 stable-spin test.

B.1 Reference Trajectories and Kinematic Baselines

To test how the forecast responds to imperfect force laws, we generated reference flights with a different drag curve,

$$C_D^{\mathrm{ref}} = 0.19 + \frac{0.20}{1+\exp[(U-20)/2.5]} + d_0, \qquad d_0 \sim \mathcal{U}(0, 0.1),$$

and a lift coefficient $0.32\tanh(4R\|\boldsymbol{\omega}\times\boldsymbol{u}\|/U^2)e^{-0.12t}$ acting along $\boldsymbol{\omega}\times\boldsymbol{u}$. Launch-velocity components were independently uniform on $[20, 36]$, $[-11, 11]$, $[2, 10]$ m/s. Wind was uniform on a horizontal disk of radius 2.5 m/s. We used equal numbers of kicks with zero spin, stable spin and a low-spin wake. Their spin-component bounds were 0, 55 and 6 rad/s. Wake flights also had a time-varying acceleration given by projecting $A(0, \sin\phi(t), 0.4\cos\phi(t))$ perpendicular to $\boldsymbol{u}$. We chose $A \in [2, 5]$ m/s^2, a period of 0.18–0.55 s and a uniform initial phase. Experiments motivate including unsteady low-spin forces [7]. The amplitudes, periods and family proportions here are synthetic.

The original study used DOP853 [5] and split whole kicks into 9 for development, 36 for calibration and 48 for blind testing. Follow-up tests through Z09 used new seeds with the same reference force laws and a 1 ms RK4 step. Z10 used DOP853 for its new reference flights. On the original 48 DOP853 flights, RK4 reproduced every first event, with maximum differences of 1.9×10^{-6} m at the crossing and 6.3×10^{-7} s in event time. We gave each predictor only its designated early frames and saved and hashed its forecasts before scoring. These comparisons test numerical agreement and performance on the simulated kick population.

On true direct entries we report Euclidean point error, region hits and mean region area. We score the probability of direct entry across first events with the Brier score $(P(G) - \mathbf{1}_G)^2$. We

compare against a least-squares fit under gravity alone and a constant-acceleration fit through the three positions. We count missing event forecasts separately from missing conditional points.

B.2 Frame Comparison at a Fixed Sampling Budget

Of the original 48 blind-test kicks, 33 entered directly, ten crossed outside the opening and five hit the ground first. The two-frame method had median point error 0.133 m and raw-region coverage 30/33, using eight times the single-frame proposal budget. We revisited the same kicks with 2,048 proposals per model for each input choice and three paired seeds (Table 11).

Table 11. Retrospective test at 100 ms with equal proposal budgets. The 33 direct entries and three seeds give 99 point forecasts. Both methods returned event estimates on all 144 attempts. Regions use raw probabilities.

Observed frames	Mean error (m)	Hits	Area (m^2)	ESS < 25
100 ms only	0.215	99/99	3.071	0
50 and 100 ms	0.170	91/99	0.288	11

At equal budget, the narrower two-frame regions covered fewer crossings. We averaged paired differences over seeds within each kick, then resampled the 33 kicks 2,000 times. The mean point-error difference (both frames minus the late frame) was -4.46 cm, with a descriptive 95% bootstrap interval of $[-9.14, 0.08]$ cm. The interval includes zero.

B.3 Tracking Noise and Coordinate Rounding

We applied three likelihoods to identical noisy positions on a separate set of 36 calibration and 48 test kicks. Before centimetre rounding we added an 8 mm Gaussian bias shared across frames for each axis and independent 6 mm Gaussian frame errors. All aerodynamic forecasts used 8,192 proposals per model and two paired seeds. One likelihood treated the combined 10 mm marginal error as independent, one used the simulated 6/8 mm decomposition, and one allowed only rounding. The two tracking models used a local velocity-proposal half-width of $0.026\,\mathrm{m}/t$. The rounding-only model used $0.007\,\mathrm{m}/t$. Importance weights corrected each proposal. The finite-budget comparison therefore includes both the likelihood and the proposal width.

Table 12. Noisy-input tests at 100 ms on 36 direct entries and two seeds. n_Q counts available point forecasts. Region hits and mean area use all 72 attempts. An undefined map gives an empty raw region. A calibrated region can encompass the whole opening.

Likelihood	Mean error (m)	n_Q	Raw hits	Raw area (m^2)	Calibrated hits	Cal. area (m^2)
Rounding only	0.425	68	27/72	0.355	72/72	17.861
Independent error	0.231	72	72/72	1.870	68/72	1.261
Shared + independent	0.257	72	71/72	1.256	71/72	1.609

For this noise test we fitted the 90% prefix-mass cutoff separately for each method and seed, using Eq. (11) and the rank rule in Sec. 6. A cutoff of one admits the entire opening, even when the raw map is empty.

The rounding-only method covered all 72/72 entries after calibration by returning the entire 17.861 m^2 goal. The shared-bias likelihood reduced mean point error relative to rounding alone by 15.86 cm on 34 complete paired kicks (95% shot-bootstrap interval 4.95–26.53 cm). Its interval against the independent-error model includes zero. Gravity had lower mean point error (0.220 m)

than either tracking model, but its event Brier score was higher (0.104 versus 0.069 for independent errors and 0.064 for shared bias). The rounding-only Brier score was 0.101 on 94/96 attempts. The other methods supplied all 96 event estimates.

B.4 Subgroup Coverage and Runtime at 100 ms

In Z09, mean area includes all 71 direct entries among 120 test kicks. At a 100% calibration target the prediction region is the entire goal. Tables 13–15 report the subgroup counts, coverage–area trade-off and computational cost for this same test.

We classify a crossing as near-edge if its distance to the nearest of the four goal edges is below one nominal cell width (0.244 m). The generator family labels are used only to score subgroups. Each kick contributed one inference seed per measurement condition. We matched differences by kick and resampled 4,000 times, using all kicks for event scores and direct entries for point error, area and hits. These intervals describe the synthetic mixture, balanced by family before the outcomes were known.

Table 13. Hits at the 90% target, by generator family and distance to the edge of the opening. Each grouping partitions the 71 entries. All subgroups use the threshold fitted for that method and measurement condition.

Profile	Stratum	Entries	100 ms	50 + 100 ms	Gravity
rounded	no spin	24	24/24	24/24	24/24
rounded	stable spin	23	16/23	21/23	22/23
rounded	wake	24	24/24	22/24	24/24
rounded	near edge	11	9/11	10/11	11/11
rounded	interior	60	55/60	57/60	59/60
tracking	no spin	24	24/24	24/24	24/24
tracking	stable spin	23	14/23	12/23	21/23
tracking	wake	24	24/24	24/24	24/24
tracking	near edge	11	9/11	7/11	10/11
tracking	interior	60	53/60	53/60	59/60

Table 14. Observed hits and mean area at selected calibration targets. Both measurement conditions use the same 71 direct entries.

Profile	Forecast	Target (%)	Hits	Mean area (m^2)
rounded	100 ms only	80	53/71	1.367
rounded	100 ms only	90	64/71	2.003
rounded	100 ms only	95	70/71	3.482
rounded	100 ms only	100	71/71	17.861
rounded	50 + 100 ms	80	58/71	0.187
rounded	50 + 100 ms	90	67/71	0.381
rounded	50 + 100 ms	95	70/71	0.585
rounded	50 + 100 ms	100	71/71	17.861
rounded	Gravity + radius	80	68/71	0.917
rounded	Gravity + radius	90	70/71	1.222
rounded	Gravity + radius	95	71/71	2.296
rounded	Gravity + radius	100	71/71	17.861
tracking	100 ms only	80	59/71	1.203
tracking	100 ms only	90	62/71	1.825
tracking	100 ms only	95	69/71	2.569
tracking	100 ms only	100	71/71	17.861
tracking	50 + 100 ms	80	54/71	0.523
tracking	50 + 100 ms	90	60/71	0.859
tracking	50 + 100 ms	95	67/71	1.385
tracking	50 + 100 ms	100	71/71	17.861
tracking	Gravity + radius	80	67/71	0.933
tracking	Gravity + radius	90	69/71	1.255
tracking	Gravity + radius	95	71/71	2.527
tracking	Gravity + radius	100	71/71	17.861

Table 15. Availability and elapsed time in one local run. Aerodynamic timings cover inference and preparation of the goal-cell masses for ranking. Gravity timings cover the ballistic solution. Both exclude application of the calibrated threshold.

Profile	Forecast	Points	Events	ESS < 25	Median (ms)	95th (ms)
rounded	100 ms only	71/71	120/120	0	90.9	103.1
rounded	50 + 100 ms	71/71	120/120	0	83.6	93.7
rounded	Gravity + radius	71/71	120/120	0	0.2	0.3
tracking	100 ms only	71/71	120/120	1	107.5	117.4
tracking	50 + 100 ms	71/71	120/120	0	121.4	130.7
tracking	Gravity + radius	71/71	120/120	0	0.2	0.3

B.5 Stable-Spin Errors under Tracking Noise

In Z10, both the nominal aerodynamic forecast and gravity gave points and regions for all 368 direct entries. With tracking errors, their Brier scores for direct entry were 0.064 and 0.093 over all 600 test kicks. Among the 110 stable-spin entries, the 95% Wilson intervals for coverage were 73.6–87.9% for aerodynamics and 60.9–77.8% for gravity. We calculated paired 95% intervals by resampling whole kicks 4,000 times with the calibration set fixed. The generator balanced the families before flight. Restricting comparisons to direct entries can leave different launch conditions in each family.

B.5.1 Acceleration Estimates from Two Frames

For an exact release point and equally spaced observations, write $\widehat{\boldsymbol{a}} = (\boldsymbol{r}_2 - 2\boldsymbol{r}_1 + \boldsymbol{r}_0)/h^2$, with $h = 0.050$ s. Approximating rounding to step q by independent uniform errors gives the per-axis variance

$$\sigma_a^2 = \frac{5(\sigma_e^2 + q^2/12) + \sigma_b^2}{h^4}.$$

The independent errors have squared coefficients $1 + 4 = 5$, and the shared frame bias has coefficient $1 - 2 = -1$. The resulting standard deviation is 2.58 m/s^2 for centimetre rounding alone and 6.76 m/s^2 with the added tracking errors. Across all 200 stable-spin kicks, the observed root-mean-square errors in the second difference, pooled over the y, z components, were 2.56 and 6.47 m/s^2. The aerodynamic point forecast on the 110 direct entries improved by 16.3 cm on average (95% interval 12.7–20.0 cm) when using rounding alone, with a likelihood and proposal width matched to each observation condition.

With release and the noisy 100 ms position fixed, a change $\delta\boldsymbol{a}$ in the second difference changes a constant-acceleration trajectory at a fixed time t by $t(t - 2h)\,\delta\boldsymbol{a}/2$. At 0.40 s, 1 m/s^2 corresponds to 6 cm. We substituted the exact simulated second differences on *all three axes* and solved again for the first plane crossing or ground contact. The forward acceleration can change the crossing time. Of the 110 true entries, 104 had plane crossings in both fits. For these the mean crossing error fell by 45.2 cm (95% interval 38.4–52.8 cm). Five had a crossing only after substitution and one only before. This sensitivity calculation requires exact positions available to the simulator.

B.5.2 Lift Priors and Recalibration

We multiplied the component half-widths of both $\boldsymbol{\beta}$ in S and $\boldsymbol{\omega}$ in E by 0.5 or 1.5. A separate test changed only the D/S/E prior weights from 0.20/0.40/0.40 to 0.02/0.49/0.49. Table 16 shows which effects survive when every prior uses the same threshold. Halving both ranges improved the true cell's mean rank from 11.92 to 6.31 and reduced mean point error by 4.17 cm (95% interval 2.15–6.28 cm). After refitting the cutoff, the corresponding hit-rate difference was +3.6 percentage points (95% interval −3.6 to +10.9). This joint change cannot separate S from E; Sec. 6.8 reports the separate reductions on the same test kicks.

B.5.3 Sensitivity to Calibration Thresholds

The stable-spin calibration entries gave the largest family cutoff (0.9698, rank 51 of 55). A rule that knows each test kick's true family could use its corresponding cutoff. The maximum rule in Fig. 8B applies 0.9698 to every kick. These rules agree on stable-spin entries. Across all 368 entries, using the maximum raised mean area from 1.003 to 1.839 m^2. We also assigned each kick to one of two

Table 16. Aerodynamic priors on the same 110 noisy stable-spin entries. The two hit columns use each prior's own pooled 90% cutoff and the nominal prior's cutoff (0.8594). Areas use each prior's own cutoff. Rank one denotes the most probable goal cell.

Prior	Hits out of 110		Mean area	Mean error	Mean rank
	Own cutoff	Nominal cutoff	(m^2)	(m)	of true cell
Nominal	90	90	1.019	0.421	11.92
Half-widths ×0.5	94	99	0.579	0.379	6.31
Half-widths ×1.5	82	68	1.315	0.452	15.25
Higher S/E weights	101	94	1.592	0.405	11.65

calibration groups by comparing the measured signal $\|(\widehat{a}_y, \widehat{a}_z + g)\|/\sigma_a$ with a prespecified boundary of 2. This rule uses only the two readings. Relative to pooling, it gained seven stable-spin hits and lost seventeen, covering 80/110 in 1.001 m^2 on average.

In the post hoc threshold exchange, the 55 pooled Z09 calibration entries gave cutoffs of 0.8214 for aerodynamic prefix mass and 0.0800 for the gravity distance score. The 180 pooled Z10 entries gave 0.8594 and 0.0687, respectively. The aerodynamic cutoff rose, admitting more cells, and the gravity radius shrank. Table 6 gives the resulting counts and areas for both test cohorts with each forecast held fixed.

Appendix C Forecast Algorithm and Physical Checks

C.1 First-Event Probability Calculation

The inputs are the release point, observed positions and times, the camera-error model and the parameter ranges. Each observation must have a valid time index and finite coordinates.

1. Draw wind and aerodynamic parameters for each force law. Propose launch velocities by inverting the latest observed position, mixing in 2% draws from the launch prior to retain support elsewhere. With only the release point, sample directly from that prior.

2. Propagate each trajectory through the available observations. A trajectory that reaches the goal plane or touches down before the last reading receives zero weight. Use the observation likelihood at all supplied times in Eq. (5). If release timing is uncertain, divide the fixed budget among quadrature nodes and combine posterior masses with their quadrature weights.

3. Propagate the surviving trajectories to the first forward crossing, descending ground contact or time horizon. Normalize each event mass by the total weight. A crossing in the open goal rectangle contributes to its goal cell. Divide each cell mass by the total direct-entry weight.

4. Return the event probabilities, modal cell, raw region and effective sample size. Zero total weight leaves the event distribution undefined. If only the direct-entry weight is zero, the event probabilities remain available and the conditional point and map are undefined.

Report the unconditional entry probability alongside the conditional point, especially near the goal boundary where entry mass may be small. Geometric bounds can exclude a flight under the stated assumptions. A low effective sample size flags uncertainty in the numerical approximation.

C.2 Lower Bound on Transverse Force

Put $L = 2h$, $\boldsymbol{a}_* = (\boldsymbol{p}_2 - 2\boldsymbol{p}_1)/h^2$ and let $\boldsymbol{n}$ be the left unit normal to the chord $\boldsymbol{p}_2 - L\boldsymbol{w}$ in the air frame. Integrating by parts gives $\boldsymbol{n} \cdot \boldsymbol{a}_* = \int_0^L \min(t, L-t)\, \boldsymbol{n} \cdot \ddot{\boldsymbol{p}}(t)\, dt/h^2$. The second difference therefore measures a time-weighted net acceleration. To minimize its inward magnitude over a wind disk of radius W, rotate the chord by the largest possible angle to the left. Its heading is then $\arg(\boldsymbol{p}_2) + \arcsin(LW/\|\boldsymbol{p}_2\|)$, with $\boldsymbol{n} \cdot \boldsymbol{p}_2 = -LW$. For these positive coordinates and negative acceleration components the inward projection decreases as this angle increases. Over the rounding box it is smallest when x_1 is largest, y_1 smallest, x_2 smallest and y_2 largest. The limiting points in the closed box are $(1.505, 0.445)$ and $(2.955, 0.865)$ m.

To separate the non-drag force from the net acceleration, write $d(t) = \boldsymbol{n} \cdot [\boldsymbol{p}(t) - t\boldsymbol{w}]$, $A = -\boldsymbol{n} \cdot \boldsymbol{a}_* = 2d(h)/h^2$ and $b(t) = At(L-t)/2$. The horizontal equation is $d'' + c(t)d' = \ell_n(t)$, with c the drag rate and ℓ_n the projected non-drag acceleration. The difference $d - b$ vanishes at $0, h, L$. Rolle's theorem applied first to $d-b$, then to $e^{\int_0^t c}(d-b)'$, gives a time s such that $-\ell_n(s) = A[1 - c(s)(h-s)]$. Over the first 100 ms, $\dot{U} \le g$ gives $U \le \|(38, 12, 11)\| + 3 + 0.1g < 45.33$ m/s. With the coefficients from Sec. 2.1, $c \le \max(\kappa_{\max}, KC_{D,\max})U < 1.854$ s^{-1}. The peak inward non-drag component must therefore reach at least $(1 - 1.854h)A$, about 8.09 m/s^2 at the displayed centres and 0.97 m/s^2 allowing rounding.

C.3 Model Assumptions and Solver Checks

Table 17. Inputs used in the nominal forecast and in the tracking-error test. A change to an assumed range or error model requires a new forecast.

Quantity	Role	Interpretation
Release, frame times and opening	Given by the problem	Fix the observations and geometry
Ball mass, radius and air density	Nominal physical inputs	Set the factor $\rho\pi R^2/(2m)$
Empirical drag curve	Literature-based fit	Limited speed range, with clipping and an offset prior
Wind and lift or spin bounds	Scenario choices	Restrict admissible flights and affect model weights
5 mm coordinate half-width	Rounding assumption	Links continuous positions to displayed readings
Cell width and RK4 steps	Numerical choices	Affect the modal cell and integration accuracy
6 mm noise and 8 mm bias	Synthetic test settings	Specify independent and shared tracking errors

Under gravity alone, $x(t) = v_{0x}t$, $y(t) = v_{0y}t$ and $z(t) = R + v_{0z}t - gt^2/2$. The analytic plane and ground times check event ordering. In still air, drag alone preserves horizontal heading because the horizontal acceleration is proportional to the horizontal velocity. Equation (2) extends this property to the frame of a constant wind. These limits check the geometry and force signs independently of posterior sampling.

Four RK4 comparisons with adaptive DOP853 cover vacuum flight, drag with ground contact, effective transverse lift and the empirical force law. The solvers agreed to within 0.171 mm in event position and 0.038 ms in event time on these cases. The original exact-centre pair in Sec. 4 also has a MATLAB replay. Independent adaptive integrations verify the two rounded-coordinate witnesses. The observations still limit what can be inferred about wind, spin and release time.

Appendix D Timing of Additional Observations

D.1 Observation Timing and Frame Count

Using Z06's 36 calibration and 48 test kicks with rounding alone, we compared three sets of observations at 8,192 proposals per model and two paired seeds. Moving from the 50/100 ms pair to the 100/150 ms pair delays the decision. Comparing that later pair with all three frames holds the 150 ms deadline fixed.

Table 18. Additional observations on 36 direct entries with two seeds (72 attempts). n_Q counts available point forecasts. An undefined map contributes a miss and zero area. Missing event estimates and low ESS use all 96 attempts.

Frames (ms)	Mean error (m)	n_Q	Hits	Area (m^2)	No event estimate	ESS < 25
50, 100	0.182	72	61/72	0.346	0	0
100, 150	0.126	72	62/72	0.136	0	20
50, 100, 150	0.130	65	52/72	0.093	6	46

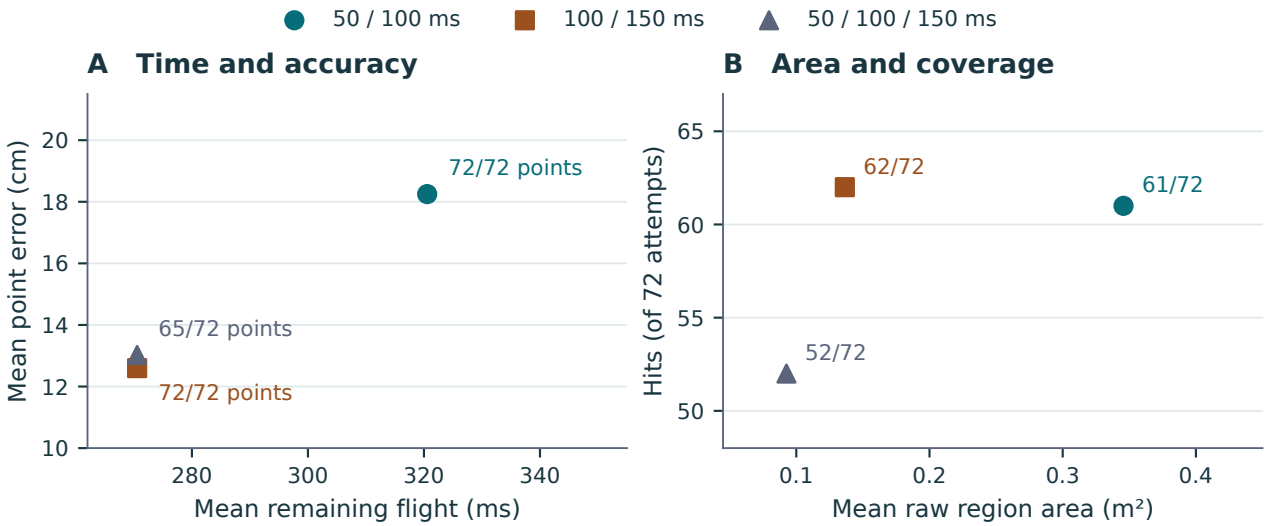


Figure 11. Forecast accuracy and time left to act. A shows mean point error for available direct-entry estimates (65/72 for three frames) and mean remaining flight time over all 36 direct entries, before computation or goalkeeper movement. B shows raw region area and coverage, with undefined maps counted as empty (Table 18).

Waiting for the later pair reduced mean point error by 5.65 cm on 36 complete paired kicks (95% shot-bootstrap interval 2.79–8.98 cm). Mean time remaining fell from 321 to 271 ms. Calibrated coverage rose from 61/72 to 64/72 while mean area fell from 0.350 to 0.177 m^2.

At the same 150 ms deadline, keeping all three frames left six of 96 attempts without an event estimate and 46/96 with low ESS. Both seeds yielded paired points on only 31 kicks. On those kicks the three-frame mean point error exceeded the later pair's by 0.27 cm (95% interval $[-1.71, 2.28]$ cm). The three-frame raw regions were smaller but covered only 52/72 entries. Few proposals fit all three readings. This limits what the comparison can establish about the earliest frame's information content. Model error and finite-test variation may also contribute. An adaptive or sequential sampler would need its own validation.

D.2 Trajectory Separation at Later Times

At 150, 200 and 250 ms, the largest coordinate gaps in the exact-centre pair (Fig. 4C) are 1.35, 5.30 and 12.96 mm. Deterministic centimetre rounding already gives different heights at 200 ms, 1.26 and 1.25 m. If each coordinate can carry any error within ±5 mm, the two reading boxes still overlap at that time. Of the three times tested, only 250 ms distinguishes the flights for every allowed error.

D.3 Frame Availability and Latency

We simulated 13 camera and noise conditions on a separate set of 48 kicks, 16 per force family and 33 direct entries. At each deadline on the estimated release clock (50 and 100 ms), we selected the latest distinct frame already received and at most 10 ms old. The predictor used its native timestamp. With 2 ms delivery delay, rates of 25, 60, 120 and 240 Hz supplied both readings on 0, 22, 47 and 48 kicks, respectively. At 240 Hz, increasing the delay to 6 ms reduced availability to 46/48. These counts depend on the simulated phase and frame-age rule. Computation takes additional time.

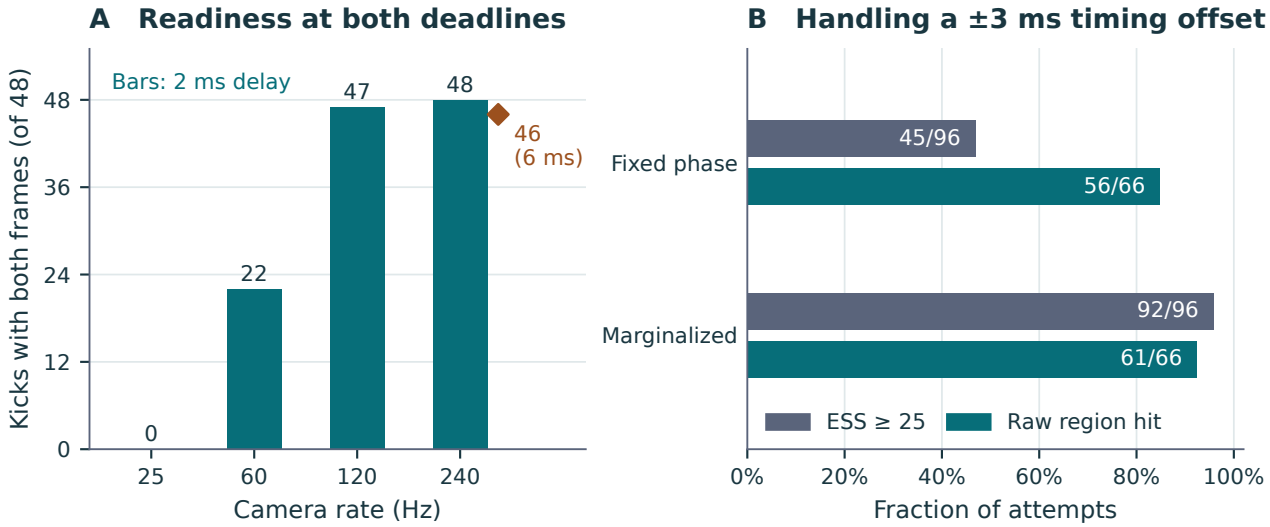


Figure 12. Timely frames and uncertain release times. A counts how many of the 48 kicks supplied both frames on time. B compares a fixed release estimate with integration over the simulated ±3 ms offset. ESS uses 96 attempts (48 kicks and two seeds). Raw-region coverage uses 66 direct-entry attempts.

For periodic frames delivered with fixed delay ℓ and no drops, the oldest possible latest frame has age bounded by $\ell + 1/f$, where f is the frame rate. To guarantee the 10 ms age limit at 2 ms delivery delay requires at least 125 Hz. A 240 Hz camera allows more margin, but consecutive exposures alone bracket release only to a half-width of about 2.08 ms. A 1 ms timing target would require an independent, finer time reference.

Integrating over the same simulated ±3 ms release offset increased the number of forecasts with ESS at least 25 from 45/96 to 92/96. Raw regions covered 61/66 entries, up from 56/66. They also grew from 0.377 to 0.543 m^2 on average, and mean point error rose from 0.196 to 0.218 m. These simulated conditions provide targets for a field trial.

Appendix E Match Tracking and Requirements for Field Validation

E.1 Eligibility and Analysis Protocol

IDSSE v1 contains three penalties in seven matches, all recorded as goals [10]. Eligibility required a release time verified to within 10 ms, two positions available by the prediction cutoff and a verifiable first event. None of the three had both release and first-event verification. We therefore use them only to diagnose the tracking assumptions.

For the retrospective check we shifted the longitudinal coordinate by 41.50 m and treated the first live frame after a dead-ball interval as an exact anchor. Regensburg's anchor was about 7 cm horizontally from the penalty spot; the other two were already 0.71 and 1.47 m downfield. Forecasts at +80 ms used the original priors, a 5 mm coordinate bound, and either the +80 ms frame alone or both the +40 and +80 ms frames. Crossing references came from interpolation of later frames in the same tracking feed, at least 0.32 s after the cutoff. We locked the forecasts before scoring, having seen the crossings during screening.

Table 19. Three penalties on the match clock. The tracked crossing is the reference. The displacement or second difference in the third column yields the required minimum acceleration under the launch assumptions. Here $\Delta^2 x = x_2 - 2x_1 + x_0$.

Fixture	Reference (Y, Z) (m)	Observed quantity (m)	Min. accel. (m/s^2)
Cologne–Bayern	$(2.750, 0.160)$	$z_1 - z_0 = -0.030$	106.25
Düsseldorf–Regensburg	$(2.919, 1.179)$	$\Delta^2 x = 0.510$	309.38
Düsseldorf–Kaiserslautern	$(3.070, 0.466)$	$\Delta^2 x = 1.100$	678.13

E.2 Acceleration Bounds between Frames

In constant wind the energy balance gives $\dot{U} \le g$, so $U \le \sqrt{38^2 + 12^2 + 11^2} + 3 + 9.81(0.080) = 45.125$ m/s over the first 80 ms. Drag and lift are orthogonal in Eq. (3). Combining them in quadrature and adding g gives upper acceleration bounds of 58.681, 89.170 and 102.648 m/s^2 for D, S and E, using $C_D \le 0.771$ and $C_L \le 0.38$ in E. Thus $A_* = 102.648$ m/s^2 bounds all three models.

For three coordinates q_0, q_1, q_2 at spacing $h = 0.040$ s, the true second difference integrates acceleration against a triangular kernel of area h^2. An exact anchor and errors bounded by ϵ in the two later positions therefore require

$$|q_2 - 2q_1 + q_0| \le A_* h^2 + 3\epsilon, \qquad z_1 - z_0 \ge v_{z,\min} h - \tfrac{1}{2} A_* h^2 - \epsilon. \tag{17}$$

The second inequality uses the launch prior $v_{z,\min} = 1.5$ m/s. Regensburg's successive x-increments of 0.71 and 1.22 m require at least $(0.510 - 0.015)/h^2 = 309.38$ m/s^2. Kaiserslautern's record repeats the anchor and then advances 1.10 m. Cologne misses the vertical bound by just 2.9 mm. All three exceed the common acceleration cap under the assumed launch conditions (Table 19).

Treating the anchor as a ball already in flight helps distinguish these failures. Relaxing the lower bound on vertical velocity from +1.5 to −11 m/s gives Cologne two forecasts with ESS 58 and 48. The other two records remain unsupported. Cologne is sensitive to the assumed initial velocity, and its release time remains unknown.

For scale, if only the middle frame were mistimed and the longitudinal speed were fixed at 30 m/s, its clock shift δ would produce $\Delta^2 x = -2 v_x \delta$. The observed second differences of +0.51 and

+1.10 m would correspond to middle frames 8.5 and 18.3 ms early. Moving all three time labels together leaves the spatial second difference unchanged. A repeated position, a mid-flight anchor or a tracking error can produce incompatible readings as well. The records do not identify a clock offset.

E.3 Single-Frame Forecast

Using only the +80 ms position, neither seed gave positive weight for Cologne or Kaiserslautern. Both placed the Regensburg modal cell at (2.074, 1.098) m, 0.849 m from the reference. The first raw region covered 4.941 m^2 and contained the reference. The second covered 4.584 m^2. With the same late frame, gravity and straight-line fits missed by 0.809 and 1.219 m. With both frames, gravity missed by 0.853 m at Regensburg and predicted ground contact first in the other two cases.

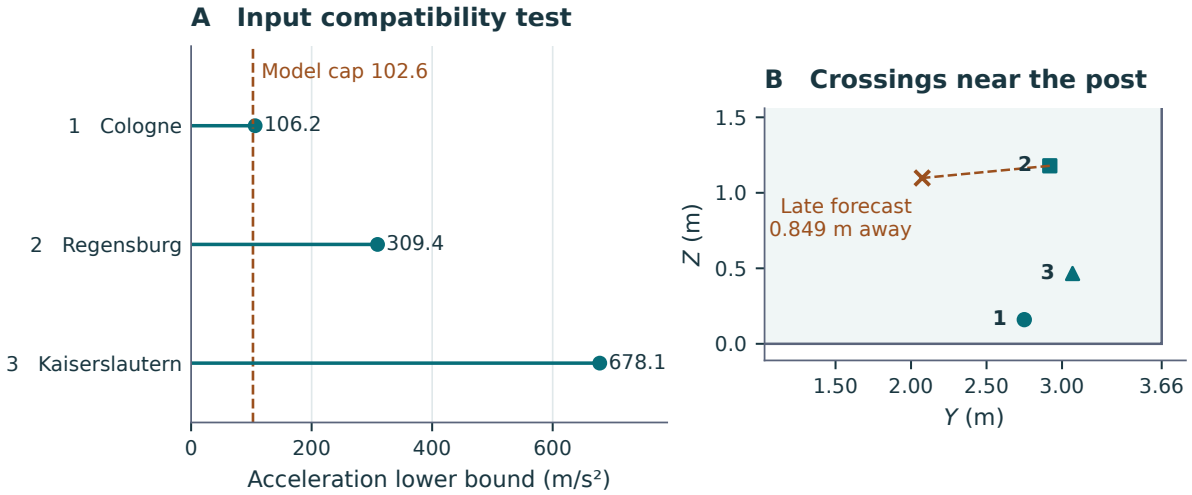


Figure 13. Tracking diagnostics for three match penalties. A compares the acceleration lower bounds under the launch assumptions with the model cap. B plots later crossings from the same tracking feed. The cross shows one +80 ms single-frame aerodynamic forecast, 0.849 m from its reference.

E.4 Independent Data for Field Validation

In ISSIA-3D [11], 47 of 79 ordinary-play segments supplied two positions 80 ms apart and a third 120 ms later. Median three-dimensional error was 0.162 m for constant-velocity extrapolation and 0.996 m for holding the last position.

A field test needs synchronized release and early-position measurements, plus an independent first-event reference. Surface markers can track spin and deflection [12]. Log frame availability, processing time, wind, ball properties, goalkeeper contacts and rebounds. The flight equations omit impacts and deformation. Use separate kicks to calibrate timing and position errors. On held-out matches report forecast availability, first-event scores and, for direct entries, point error, coverage and area.

Appendix F Computational Reproducibility

The reproduction guide covers the prescribed forecast (seeds 8111 and 8117, 30×10 grid), the rounded-coordinate flights, and the timing and spin tests. The archives preserve separate calibration and test scores, thresholds, seeds and forecast hashes. The revision-controls script checks forecast locks and replays observation scenarios. Frozen summaries support rebuilding the figures and coverage tables.

References

[1] University Physics Competition, 2026 contest, Problem B. Official problem statement.

[2] J. E. Goff and M. J. Carré, Soccer ball lift coefficients via trajectory analysis. *European Journal of Physics* **31**, 775–784 (2010). DOI 10.1088/0143-0807/31/4/007.

[3] T. Asai, K. Seo, O. Kobayashi and R. Sakashita, Fundamental aerodynamics of the soccer ball. *Sports Engineering* **10** (2007). DOI 10.1007/BF02844207.

[4] J. E. Goff, J. Kelley, C. Hobson, K. Seo, T. Asai and S. Choppin, Creating drag and lift curves from soccer trajectories. *European Journal of Physics* **38**, 044003 (2017). DOI 10.1088/1361-6404/aa6fcd.

[5] P. Virtanen et al., SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods* **17**, 261–272 (2020). DOI 10.1038/s41592-019-0686-2.

[6] L. F. Shampine and M. W. Reichelt, The MATLAB ODE Suite. *SIAM Journal on Scientific Computing* **18**, 1–22 (1997). DOI 10.1137/S1064827594276424.

[7] T. Mizota et al., The strange flight behaviour of slowly spinning soccer balls. *Scientific Reports* **3**, 1871 (2013). DOI 10.1038/srep01871.

[8] International Football Association Board, Laws of the Game, Law 2: The Ball. Ball specifications, accessed 7 November 2026.

[9] J. Lei, M. G'Sell, A. Rinaldo, R. J. Tibshirani and L. Wasserman, Distribution-Free Predictive Inference for Regression. arXiv:1604.04173.

[10] M. Bassek, R. Rein, H. Weber and D. Memmert, An integrated dataset of spatiotemporal and event data in elite soccer. *Scientific Data* **12**, 195 (2025). DOI 10.1038/s41597-025-04505-y. IDSSE v1 data, CC BY 4.0, DOI 10.6084/m9.figshare.28196177.v1.

[11] M. Gutiérrez-Pérez and A. Agudo, SoccerNet-v3D: Leveraging sports broadcast replays for 3D scene understanding. *IEEE/CVF CVPR Workshops* (2025). DOI 10.1109/CVPRW67362.2025.00595. ISSIA-3D annotations, release 1.0.0.

[12] I. Griffiths, C. Evans and N. Griffiths, Tracking the flight of a spinning football in three dimensions. *Measurement Science and Technology* **16**, 2056–2065 (2005). DOI 10.1088/0957-0233/16/10/022.

Report on the Use of AI

Literature review. We used OpenAI Codex to locate studies of football aerodynamics and open ball-tracking datasets, compare their measurement methods, and identify evidence relevant to our force models and field tests.

Experimental code. Codex helped write and debug Python scripts for trajectory integration, blind testing on synthetic kicks, and calibration of prediction regions. It also helped implement independent trajectory checks in MATLAB and trace numerical results back to the simulation outputs.

Manuscript revision. Codex helped draft the English manuscript from the experimental report and refine its wording, headings, figures, and captions. In particular, it helped distinguish the model probability for one kick from the coverage measured across independent kicks.